Bulk–Boundary Decomposition for Adaptive Windows in Spectral/HAC Estimation: Mechanism, Scale, and Certified Negativity

March 6, 2026

Abstract

Adaptive bandwidth choices in HAC and spectral estimation are often implemented through state-dependent truncations. While classical tapered lag windows deliver nonnegative spectral windows and hence positive semidefinite (PSD) covariance estimators, hard cutoffs can generate boundary ringing and negative estimated spectra in finite samples. We provide an explicit diagnostic of this boundary channel using a fully tractable oscillatory-integral analogue with a moving truncation boundary. An integration-by-parts boundary decomposition isolates a sine-transform boundary term and an interior remainder, and yields a sufficient defect-density condition for nonnegativity. For a two-parameter family of strictly positive kernels we certify a sign reversal of the associated cosine transform using explicit, reproducible truncation and quadrature error bounds. Complementary Monte Carlo evidence shows that, under a common state-dependent bandwidth path, hard truncation exhibits non-PSD failures while a Bartlett taper eliminates them by construction. The results clarify PSD failures as boundary artifacts and motivate operational diagnostics and design rules for adaptive windowing.

Keywords: HAC covariance estimation; long-run variance; spectral density estimation; adaptive bandwidth; state-dependent truncation; spectral windows; positive semidefinite constraints; Dirichlet and Fejér kernels; numerical error bounds.

C12; C13; C22

MSC 2020: 62M10 62F12 65G40 65D32 42A38

1 Introduction

Nonnegativity of the spectral density and positive semidefiniteness (PSD) of the associated covariance estimator are coherence requirements in spectral estimation and HAC covariance construction Andrews, 1991; Brockwell and Davis, 1991; Newey and West, 1987; Priestley, 1981. In practice, however, negative spectral estimates and non-PSD covariance matrices can arise under modern *adaptive* implementations, where tuning choices (e.g., window length, bandwidth, or truncation rules) depend on nuisance parameters, preliminary estimates, or local features of the data. This paper studies a stylized mechanism through which such pathologies can be *boundary-induced*: they can emerge from a parameter-dependent cutoff that generates oscillatory boundary contributions, even when the underlying kernel is strictly positive.

Stylized HAC mapping (why this is a spectral/HAC issue). Many adaptive (state-dependent) windowing rules can be viewed, at a stylized level, as inducing a *moving truncation*.

After rewriting an estimator as an integral (or sum) against a windowed kernel, the effective domain of integration depends on a parameter and therefore generates an explicit *boundary term*. This boundary term is not controlled by pointwise kernel positivity and can, in principle, induce negative spectral values or non-PSD covariance matrices. The analysis below isolates this boundary contribution via a bulk–boundary decomposition, explains why any induced negativity may be extremely small due to an exact cancellation (Schwartz decay), and provides a verifiable design criterion (via a defect density) to diagnose or avoid boundary-induced sign reversals under adaptive truncations.

Sharp truncations are well known to generate oscillatory artifacts in Fourier-type representations. The present paper studies an analogous boundary mechanism for a family of oscillatory integral transforms assembled from *variable spatial windows*. Here the oscillations are not driven by discontinuities of the target function, but by a *moving truncation boundary*. This observation also motivates a practical mitigation: replacing hard cutoffs by tapered kernels. In our Monte Carlo section, holding the same state-dependent bandwidth path fixed, a Bartlett taper (E3s) yields no admissibility failures in the reported runs, consistent with the interpretation that discontinuous cutoffs amplify boundary-driven ringing under the chosen state rule Bartlett, 1950. Our goal is not to claim a universal theorem for all adaptive windows; rather, we provide an explicit diagnostic showing how boundary terms can overturn naive positivity intuition in a concrete, fully tractable setting.

More concretely, we consider cosine transforms of the form

$$W_{\alpha,\beta}(t) = \int_{-\infty}^{\infty} H_{\alpha,\beta}(x)\cos(tx)\,dx, \qquad t \in \mathbb{R}, \tag{1}$$

where the kernel $H_{\alpha,\beta}$ is given by an explicit *truncated* Laplace–type representation

$$H_{\alpha,\beta}(x) = \int_{2\pi e^{|x|}}^{\infty} w(a)\,P_{\alpha,\beta}(a,x)\,e^{-a\cosh x}\,da, \qquad P_{\alpha,\beta}(a,x) = a^2 + \beta - \alpha a\cosh x. \tag{2}$$

The truncation threshold $a \geq 2\pi e^{|x|}$ depends on $|x|$ and therefore induces a *moving boundary*. After exchanging the order of integration (Tonelli/Fubini), this moving boundary becomes a variable integration window

$$|x| \leq L(a), \qquad L(a) = \log\Big(\frac{a}{2\pi}\Big),$$

so that $W_{\alpha,\beta}$ is assembled from oscillatory integrals over intervals whose length is logarithmic in a. This is precisely the configuration in which boundary ringing is expected: the oscillatory phase $\cos(tx)$ interacts with a hard cutoff at $x = \pm L(a)$, and boundary contributions appear that are invisible to pointwise estimates on $H_{\alpha,\beta}$.

The first part of the paper establishes that this boundary mechanism cannot be dismissed by kernel positivity. Indeed, for an open parameter range $(\alpha,\beta) \in (0,2\pi) \times (0,\infty)$ we prove that

$$H_{\alpha,\beta}(x) > 0 \qquad \text{for all } x \in \mathbb{R},$$

by elementary inequalities (Theorem 4.2). Thus any sign changes of $W_{\alpha,\beta}$ cannot be attributed to local negativity of the kernel, but must arise from *boundary oscillations induced by truncation*.

To keep the moving-truncation geometry explicit while separating it from routine analytic justification, Section 3 states a concise set of verifiable regularity and integrability conditions under which Tonelli/Fubini and integration by parts can be applied without introducing spurious boundary terms. These conditions are not meant to suggest generality beyond what is actually proved here:

all substantive conclusions are established for the explicit family $H_{\alpha,\beta}$, and the auxiliary conditions function only as a transparent checklist for interchange and differentiation steps.

The second part of the paper derives an explicit *bulk–boundary decomposition* that isolates the boundary mechanism. An integration–by–parts identity separates an oscillatory *boundary sine transform* generated by the moving cutoff, together with an interior remainder:

$$W_{\alpha,\beta}(t) = \frac{2}{t}\, B_{\alpha,\beta}(t) - \frac{2}{t}\, R_{\alpha,\beta}(t) \qquad (t \neq 0).$$

Here $B_{\alpha,\beta}$ is a sine transform of an explicit boundary density (coming from $x = L(a)$), while $R_{\alpha,\beta}$ is an interior term. This shows that global sign control for $W_{\alpha,\beta}$ is necessarily a *harmonic comparison problem*: even with $H_{\alpha,\beta} > 0$, boundary ringing can dominate in certain frequency regimes.

A key structural feature is that, despite the boundary term admitting an explicit algebraic asymptotic expansion, an exact cancellation identity forces the combination $B_{\alpha,\beta} - R_{\alpha,\beta}$ to decay beyond all orders, so that $W_{\alpha,\beta}$ is a Schwartz function of t (Proposition 5.9). Consequently, when negativity occurs it can occur at very small magnitudes, a fact confirmed via validated numerics in Appendix A. The specific instance $(\alpha, \beta) = (6, 9)$ is used there only as a concrete reference instance.

Finally, we formulate a structural sufficient condition for global nonnegativity of $W_{\alpha,\beta}$ in terms of a boundary dominance inequality (Proposition 6.3). To show that the boundary positivity requirement is not vacuous, we exhibit a nontrivial class of boundary densities with everywhere nonnegative sine transform, based on Laplace mixtures (Lemma 6.2).

Organization of the paper. Section 2 motivates the problem from econometric practice (spectral coherence, HAC admissibility, and adaptive windowing) and situates it within the HAC and PSD-repair literatures. Section 3 formalizes the moving-truncation/variable-window representation (including the bookkeeping class $\mathcal{K}(C)$) and introduces the explicit two-parameter kernel family $H_{\alpha,\beta}$ together with the cosine-transform functional $W_{\alpha,\beta}$ and its continuity in (α, β). Section 4 establishes global pointwise positivity of $H_{\alpha,\beta}$ for $0 < \alpha < 2\pi$ and $\beta > 0$. Section 5 derives the bulk–boundary decomposition, makes the truncation-induced boundary term explicit (via a closed-form boundary density), and proves the exact cancellation identity and the resulting rapid (Schwartz-type) decay of $W_{\alpha,\beta}$; it also records the deterministic truncation/quadrature bounds used for numerical certification. Section 6 formulates a verifiable sufficient condition for global nonnegativity of $W_{\alpha,\beta}$ in terms of the defect density $\kappa = -H'$, and provides an explicit nontrivial positive test case based on Laplace mixtures. Section 7 reports Monte Carlo evidence for regression-based HAC estimators under fixed and state-dependent truncations and documents an empirical parameter map for sign changes in the analytic family. Section 8 concludes. Appendix A provides validated-numerics sign certification for the reference parameters $(\alpha, \beta) = (6, 9)$ and a sparse set of additional representative points.

2 Econometric background: HAC, spectral coherence, and adaptive windows

In econometrics, spectral density estimation and heteroskedasticity–autocorrelation consistent (HAC) covariance construction impose basic coherence requirements: the estimated spectral density must satisfy $f(\omega) \geq 0$ for all frequencies, and the associated covariance estimator must be positive semidefinite (PSD) Andrews, 1991; Brockwell and Davis, 1991; Newey and West, 1987; Priestley, 1981. Classical kernel-based HAC estimators achieve PSD by construction when the weighting scheme corresponds to a nonnegative spectral window Andrews, 1991; Newey and West, 1987. In applied implementations, however, bandwidth and truncation choices are often data-dependent

or state-dependent, with the effective window length depending on preliminary estimates or local features of the sample.

At a stylized level, many adaptive rules can be written as

$$\widehat{\Omega} = \sum_{|k| \le L(\theta)} K\left(\frac{k}{L(\theta)}\right) \widehat{\Gamma}(k),$$

where $L(\theta)$ is a bandwidth depending on an estimated parameter θ. When such estimators are re-expressed as integrals (or sums) against windowed kernels, parameter dependence can be interpreted as an effective moving cutoff. In continuous analogues, this corresponds to an integration domain whose boundary depends on the evaluation variable, generating explicit boundary terms under Tonelli/Fubini interchanges and integration-by-parts manipulations.

The key point is that PSD or nonnegativity is not ensured by pointwise kernel positivity alone: when the truncation boundary depends on a state variable, oscillatory boundary contributions can arise that are invisible to local kernel bounds. The analysis below isolates this mechanism in a fully explicit setting. We do not claim a general theorem for adaptive HAC estimators; rather, we provide a tractable analogue that clarifies how parameter-dependent cutoffs can, in principle, induce boundary-driven sign reversals.

2.1 Related work in HAC, adaptive bandwidths, and PSD corrections

The construction of HAC covariance estimators originates in the kernel-based long-run variance framework of Newey and West, 1987, with asymptotic MSE-optimal bandwidth selection developed in Andrews, 1991. Subsequent refinements include prewhitening and automatic bandwidth rules Andrews and Monahan, 1992, higher-order kernel designs Newey and West, 1994, and fixed-b asymptotics Kiefer and Vogelsang, 2002, 2005. In these constructions, nonnegativity of the implied spectral window is central for ensuring PSD of the resulting covariance matrix.

Data-dependent or plug-in bandwidth selection is now standard in applied work. Automatic procedures based on preliminary parameter estimation or spectral pilot estimators Andrews, 1991; Newey and West, 1994 introduce an explicit dependence of the effective truncation length on estimated quantities. More broadly, adaptive smoothing and state-dependent bandwidth ideas appear throughout spectral estimation and nonparametric time-series analysis Brockwell and Davis, 1991; Phillips et al., 2006; Politis, 2011; Priestley, 1981, where local window-length selection can be interpreted as inducing a variable integration domain.

From a practical standpoint, finite-sample non-PSD covariance estimates are sometimes corrected ex post. Common remedies include projection onto the PSD cone (e.g., eigenvalue truncation Higham, 2002) and shrinkage-based regularization Ledoit and Wolf, 2004. Such repairs restore feasibility but do not address the structural origin of potential sign reversals under adaptive truncation. The present paper complements this literature by providing a mechanism-based diagnostic: in a fully explicit oscillatory-integral analogue, we show how a parameter-dependent cutoff generates boundary sine transforms that can overturn naive positivity intuition, even when the underlying kernel is strictly positive.

3 Moving truncations, variable windows, and the explicit kernel family

This section records a generic variable-window representation for spatially truncated kernels under routine integrability and smoothness conditions, and then introduces the explicit two–parameter

family that constitutes the main object of study in this paper.

3.1 Technical setup for variable-window kernels

We collect a minimal set of analytic hypotheses needed to (i) exchange the order of integration (Tonelli/Fubini) and obtain a *variable-window* representation, and (ii) perform integration by parts in x without generating unintended boundary terms Katznelson, 2004; Rudin, 1987. These hypotheses are used only for bookkeeping: throughout, all substantive sign statements are proved for the explicit family $H_{\alpha,\beta}$ via the boundary mechanism and validated numerics.

Fix a constant $C > 0$ and set

$$L(a) := \log\Big(\frac{a}{C}\Big), \qquad a \geq C.$$

Let $w : (C, \infty) \to (0, \infty)$ be measurable and let $\{f_a\}_{a \geq C}$ be a family of even C^∞ functions $f_a : \mathbb{R} \to \mathbb{R}$. We associate to $(C, w, \{f_a\})$ the even kernel

$$H(x) := \int_C^\infty w(a)\, f_a(x)\, \mathbf{1}_{\{|x| \leq L(a)\}}\, da \;=\; \int_{Ce^{|x|}}^\infty w(a)\, f_a(x)\, da, \qquad x \in \mathbb{R}. \tag{3}$$

Standing conditions (Tonelli/IBP bookkeeping). We will invoke the following conditions when justifying Tonelli/Fubini and integration-by-parts steps.

(K1) (*Super–exponential envelope*) For every $m \geq 0$ there exist constants $c_m > 0$ and integers $p_m \geq 0$ such that

$$\big|\partial_x^m f_a(x)\big| \leq c_m (1 + a)^{p_m} e^{-a \cosh x} \qquad \text{for all } a \geq C \text{ and } |x| \leq L(a). \tag{4}$$

(K2) (*Admissible weight*) For every integer $q \geq 0$,

$$\int_C^\infty w(a)\, (1 + a)^q e^{-a}\, da < \infty. \tag{5}$$

(K3) (*Positivity on the truncated domain; used only when needed*) For every $a \geq C$,

$$f_a(x) > 0 \qquad \text{for all } |x| \leq L(a) = \log(a/C). \tag{6}$$

Remark 3.1. **(Role of the conditions).** Conditions **(K1)**–**(K2)** are used to justify Tonelli/Fubini and the integration-by-parts manipulations underlying the bulk–boundary decomposition. Condition **(K3)** is introduced only as a convenient positivity flag: once verified for a concrete family, it permits recording kernel positivity statements without re-proving the same local inequality.

(Truncation geometry). The truncation $a \geq Ce^{|x|}$ is equivalent to $|x| \leq L(a) = \log(a/C)$, hence the inner domain produced by Tonelli/Fubini is a *variable window* whose length is asymptotically logarithmic: $L(a) = \log a + O(1)$.

Lemma 3.2 (Variable-window (Tonelli/Fubini) representation)**.** *Assume that H is given by* (3) *and that **(K1)**–**(K2)** hold. Define its cosine transform*

$$W(t) := \int_{\mathbb{R}} H(x) \cos(tx)\, dx.$$

Then Tonelli/Fubini yields the variable-window *representation: the moving cutoff $a \geq Ce^{|x|}$ is equivalent to $|x| \leq L(a)$, and therefore for every $t \in \mathbb{R}$,*

$$W(t) = \int_C^\infty w(a) \left(\int_{-L(a)}^{L(a)} f_a(x) \cos(tx)\, dx \right) da, \qquad L(a) = \log(a/C). \tag{7}$$

Moreover, since f_a is even, this reduces to

$$W(t) = 2 \int_C^\infty w(a) \left(\int_0^{L(a)} f_a(x) \cos(tx)\, dx \right) da.$$

Proof. By **(K1)** with $m = 0$,

$$|f_a(x)| \mathbf{1}_{\{|x| \leq L(a)\}} \leq c_0 (1+a)^{p_0} e^{-a \cosh x} \mathbf{1}_{\{|x| \leq L(a)\}}.$$

Hence, for each $a \geq C$,

$$\int_{-L(a)}^{L(a)} |f_a(x)|\, dx \leq c_0 (1+a)^{p_0} \int_{-L(a)}^{L(a)} e^{-a \cosh x}\, dx \leq 2c_0 (1+a)^{p_0} L(a) e^{-a},$$

since $\cosh x \geq 1$ on $\mathbb{R}$. Therefore

$$\int_{\mathbb{R}} \int_C^\infty |w(a) f_a(x)| \mathbf{1}_{\{|x| \leq L(a)\}}\, da\, dx \leq 2c_0 \int_C^\infty w(a)(1+a)^{p_0} L(a) e^{-a}\, da < \infty,$$

and the last integral is finite by **(K2)** because $L(a) = \log(a/C) \ll (1+a)$. Tonelli/Fubini applies and yields (7). The even reduction is immediate. □

3.2 Definition of the explicit family

We introduce a two–parameter family of truncated kernels. Fix parameters $\alpha > 0$ and $\beta > 0$. Let $H_{\alpha,\beta} : \mathbb{R} \to \mathbb{R}$ be the even function defined by

$$H_{\alpha,\beta}(x) = \int_{2\pi e^{|x|}}^\infty w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a \cosh x}\, da, \qquad x \in \mathbb{R}, \tag{8}$$

where the weight w is given by

$$w(a) = \frac{1}{8\pi} \log\Big(\frac{a}{2\pi}\Big) \Big(\frac{a}{2\pi}\Big)^{3/2}, \qquad a \geq 2\pi, \tag{9}$$

and the polynomial prefactor is

$$P_{\alpha,\beta}(a,x) = a^2 + \beta - \alpha a \cosh x. \tag{10}$$

The truncation threshold $a \geq 2\pi e^{|x|}$ will be responsible for the boundary phenomena analyzed later.

Fix (α, β) and, for notational convenience, write

$$H(x) := H_{\alpha,\beta}(x), \qquad P(a,x) := P_{\alpha,\beta}(a,x), \qquad W(t) := W_{\alpha,\beta}(t).$$

Remark 3.3. All analytic statements in Sections 2–5 are formulated for general parameters (α, β) in the ranges explicitly stated in each result. The particular choice $(\alpha, \beta) = (6, 9)$ is used *only* for validated-numerics certification in Appendix A (and in any figure/table explicitly labeled as "reference case").

Lemma 3.4 (The explicit family satisfies **(K1)**–**(K2)**). *Fix* $(\alpha, \beta) \in (0, 2\pi) \times (0, \infty)$ *and set* $C := 2\pi$. *Define*

$$f_{a,\alpha,\beta}(x) := P_{\alpha,\beta}(a, x)e^{-a\cosh x}, \qquad P_{\alpha,\beta}(a, x) = a^2 + \beta - \alpha a \cosh x,$$

with w *as in* (9). *Then the kernel* $H_{\alpha,\beta}$ *in* (8) *satisfies the bookkeeping conditions* ***(K1)***–***(K2)*** *above (with* $C = 2\pi$*). In particular, the variable-window representation in Lemma 3.2 applies to* $W_{\alpha,\beta}$.

Proof. Differentiating $f_{a,\alpha,\beta}(x) = P_{\alpha,\beta}(a, x)e^{-a\cosh x}$ produces finite linear combinations of terms in which derivatives fall on $P_{\alpha,\beta}$ and on $e^{-a\cosh x}$. Each such term is bounded on $|x| \le L(a)$ by a constant times $(1 + a)^{p_m} e^{-a\cosh x}$ for suitable integers p_m, yielding (4).

For (5), note that $w(a)$ has polynomial–log growth, indeed $w(a) \asymp a^{3/2} \log a$ as $a \to \infty$, so $w(a)(1 + a)^q e^{-a}$ is integrable on $(2\pi, \infty)$ for every $q \ge 0$ because the exponential factor dominates any polynomial–log term. □

3.3 Basic properties

We record several elementary properties of the kernel $H_{\alpha,\beta}$ that follow directly from its definition.

Lemma 3.5. *Fix* $\alpha, \beta > 0$ *and let* $H = H_{\alpha,\beta}$ *be given by* (8). *Then:*

1. $H(x) = H(-x)$ *for all* $x \in \mathbb{R}$.
2. $H \in C^\infty(\mathbb{R})$.
3. $H \in L^1(\mathbb{R})$.

Proof. (1) Evenness. Immediate, since $\cosh x$ and $|x|$ are even.

(2) Smoothness. For $x > 0$, write

$$H(x) = \int_{2\pi e^x}^{\infty} F(a, x)\, da, \qquad F(a, x) := w(a)\, P_{\alpha,\beta}(a, x)\, e^{-a\cosh x}.$$

For each fixed a, the map $x \mapsto F(a, x)$ is C^∞, and for each $k \ge 0$, $\partial_x^k F(a, x)$ is a finite linear combination of terms of the form

$$w(a)\,(1 + a)^{q_k}\,(\cosh x)^{r_k}(\sinh x)^{s_k}\, e^{-a\cosh x}.$$

Fix $x_0 > 0$ and choose $\delta \in (0, x_0)$. For $x \in [x_0 - \delta, x_0 + \delta]$ we have $\cosh x \ge 1$ and $2\pi e^x \ge 2\pi e^{x_0-\delta} =: A_0$, hence

$$|\partial_x^k F(a, x)| \le C_k\, w(a)\,(1 + a)^{q_k}\, e^{-a} \quad \text{for all } a \ge A_0,\ x \in [x_0 - \delta, x_0 + \delta].$$

The right-hand side is integrable in a since $w(a)$ has at most polynomial–log growth and e^{-a} dominates. Therefore the standard Leibniz rule and dominated convergence justify differentiation under the integral sign on $(0, \infty)$, including the boundary contribution induced by the moving lower limit $a = 2\pi e^x$. Iterating yields $H \in C^\infty((0, \infty))$, and evenness gives $H \in C^\infty(\mathbb{R})$.

(3) Integrability. The moving lower limit implies a super–exponentially small tail in $|x|$. On the integration domain we have $\cosh x \ge \frac{1}{2}e^{|x|}$ and $a \ge 2\pi e^{|x|}$, hence

$$a\cosh x \;\ge\; \tfrac{1}{2}\, a e^{|x|} \;\ge\; \tfrac{1}{2}(2\pi e^{|x|})e^{|x|} \;=\; \pi e^{2|x|}.$$

Using $|P_{\alpha,\beta}(a,x)| \le C(a^2+1)$ and $w(a) \lesssim a^{3/2}\log a$ for large a, we obtain for some integer q and constants C', C'',

$$|H(x)| \le \int_{2\pi e^{|x|}}^{\infty} C\, a^q e^{-a\cosh x}\, da \le C'\, e^{-\pi e^{2|x|}} \int_{2\pi e^{|x|}}^{\infty} a^q\, da \le C''\, e^{-\pi e^{2|x|}} (1+e^{|x|})^{q+1}.$$

In particular, $|H(x)| \le C_m e^{-m|x|}$ for every $m > 0$ and sufficiently large $|x|$, which implies $H \in L^1(\mathbb{R})$. □

All changes of order of integration and differentiation under the integral sign performed in this paper are justified by Tonelli's theorem and dominated convergence, using envelope bounds of the form

$$|w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}| \le C\, w(a)(a^2+1)e^{-a\cosh x},$$

which are integrable in a uniformly in $x \in \mathbb{R}$ on the domain $a \ge 2\pi e^{|x|}$. In particular,

$$\int_{\mathbb{R}} \int_{2\pi e^{|x|}}^{\infty} |w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}|\, da\, dx < \infty,$$

so Tonelli/Fubini and differentiation under the integral sign apply throughout the paper.

3.4 The cosine–transform functional

We define the cosine–transform functional associated with the kernel H by

$$W(t) = \int_{-\infty}^{\infty} H(x)\, \cos(tx)\, dx, \qquad t \in \mathbb{R}. \tag{11}$$

By Lemma 3.5, the integral in (11) converges absolutely for all $t \in \mathbb{R}$, and W is continuous and even.

The purpose of the present paper is to analyze how the explicit structure of H, and in particular the moving truncation encoded by the lower limit in (8), influences the sign of the cosine–transform functional $W(t)$.

3.5 Dependence on the parameters and robustness of sign changes

For $\alpha, \beta > 0$, define the cosine–transform functional associated with $H_{\alpha,\beta}$ by

$$W_{\alpha,\beta}(t) = \int_{-\infty}^{\infty} H_{\alpha,\beta}(x) \cos(tx)\, dx, \qquad t \in \mathbb{R}. \tag{12}$$

In particular, under the shorthand introduced above, $W(t) = W_{\alpha,\beta}(t)$.

Lemma 3.6 (Continuity in (α,β))**.** *Fix $t \in \mathbb{R}$. The map $(\alpha,\beta) \mapsto W_{\alpha,\beta}(t)$ is continuous on $(0,2\pi) \times (0,\infty)$. More precisely, if $(\alpha_n,\beta_n) \to (\alpha,\beta)$ with $0 < \alpha_n < 2\pi$ and $\beta_n > 0$, then*

$$W_{\alpha_n,\beta_n}(t) \to W_{\alpha,\beta}(t).$$

Proof. Using (8) and (12), Tonelli/Fubini yields the absolutely convergent representation

$$W_{\alpha,\beta}(t) = \int_{\mathbb{R}} \int_{2\pi e^{|x|}}^{\infty} w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}\, \cos(tx)\, da\, dx,$$

where $P_{\alpha,\beta}(a,x) = a^2 + \beta - \alpha a \cosh x$. Let $(\alpha_n, \beta_n) \to (\alpha, \beta)$. After restricting to n sufficiently large, we may assume

$$\alpha^* := \sup_n \alpha_n < 2\pi, \qquad \beta^* := \sup_n \beta_n < \infty.$$

On the domain $a \geq 2\pi e^{|x|}$ we have $\cosh x \leq e^{|x|} \leq a/(2\pi)$, hence

$$|P_{\alpha_n,\beta_n}(a,x)| \leq a^2 + \beta^* + \alpha^* a \cosh x \leq a^2 + \beta^* + \frac{\alpha^*}{2\pi} a^2 \leq C\,(a^2+1),$$

for a constant C independent of n, a, and x. Therefore,

$$\left| w(a)\, P_{\alpha_n,\beta_n}(a,x)\, e^{-a\cosh x} \cos(tx) \right| \leq C\, w(a)\, (a^2+1)\, e^{-a\cosh x}.$$

The right-hand side is integrable on $\mathbb{R} \times (2\pi, \infty)$ by the same absolute-integrability bounds used earlier to justify Tonelli/Fubini. Since the integrand converges pointwise as $n \to \infty$, dominated convergence implies $W_{\alpha_n,\beta_n}(t) \to W_{\alpha,\beta}(t)$. □

Corollary 3.7 (Robustness of negativity)**.** *Let $(\alpha_0, \beta_0) \in (0, 2\pi) \times (0, \infty)$ and suppose there exists $t_0 \in \mathbb{R}$ such that $W_{\alpha_0,\beta_0}(t_0) < 0$. Then there exists $\delta > 0$ such that*

$$W_{\alpha,\beta}(t_0) < 0 \qquad \textit{whenever} \qquad |\alpha - \alpha_0| + |\beta - \beta_0| < \delta.$$

In particular, once a negative value is certified for the reference parameters $(\alpha, \beta) = (6, 9)$, the sign-change phenomenon persists for an open set of nearby parameters.

Proof. Set $\eta := -\frac{1}{2} W_{\alpha_0,\beta_0}(t_0) > 0$. By Lemma 3.6, there exists $\delta > 0$ such that $|W_{\alpha,\beta}(t_0) - W_{\alpha_0,\beta_0}(t_0)| < \eta$ whenever $|\alpha - \alpha_0| + |\beta - \beta_0| < \delta$. Hence

$$W_{\alpha,\beta}(t_0) < W_{\alpha_0,\beta_0}(t_0) + \eta = \tfrac{1}{2} W_{\alpha_0,\beta_0}(t_0) < 0,$$

which proves the claim. □

4 Global pointwise positivity of the kernel

In this section we formalize a simple positivity bookkeeping step for truncated-kernel representations and then verify the positivity condition **(K3)** for the explicit family introduced in Section 3.

Remark 4.1 (Bookkeeping: **(K3)** implies pointwise positivity). Suppose H admits a truncated-kernel representation of the form

$$H(x) = \int_{Ce^{|x|}}^{\infty} w(a)\, f_a(x)\, da, \qquad x \in \mathbb{R},$$

with $w(a) > 0$ for a.e. $a \in [C, \infty)$. If **(K3)** holds, i.e. if $f_a(x) > 0$ on the integration domain $\{|x| \leq \log(a/C)\}$, then $H(x) > 0$ for all $x \in \mathbb{R}$.

4.1 Kernel positivity for the explicit family (preliminary check)

We recall that the kernel $H_{\alpha,\beta}$ admits the representation

$$H_{\alpha,\beta}(x) = \int_{2\pi e^{|x|}}^{\infty} w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}\, da,$$

where the weight $w(a)$ is strictly positive for $a > 2\pi$ and

$$P_{\alpha,\beta}(a,x) = a^2 + \beta - \alpha a \cosh x.$$

Theorem 4.2 (Verification of **(K3)** for the explicit family)**.** *Fix (α, β) with $0 < \alpha < 2\pi$ and $\beta > 0$. Then for every $a \geq 2\pi$ one has*

$$f_{a,\alpha,\beta}(x) = P_{\alpha,\beta}(a, x)e^{-a\cosh x} > 0 \qquad \textit{for all } |x| \leq \log(a/2\pi).$$

Consequently, $H_{\alpha,\beta}(x) > 0$ for all $x \in \mathbb{R}$.

Proof. Fix $a \geq 2\pi$ and let $|x| \leq \log(a/2\pi)$, equivalently $a \geq 2\pi e^{|x|}$. Using $\cosh x \leq e^{|x|}$ we have

$$\cosh x \leq e^{|x|} \leq \frac{a}{2\pi},$$

and therefore

$$\alpha a \cosh x \leq \alpha a \cdot \frac{a}{2\pi} = \frac{\alpha}{2\pi}\, a^2.$$

It follows that

$$P_{\alpha,\beta}(a, x) = a^2 + \beta - \alpha a \cosh x \geq a^2 \left(1 - \frac{\alpha}{2\pi}\right) + \beta > 0,$$

since $0 < \alpha < 2\pi$ and $\beta > 0$. Hence $f_{a,\alpha,\beta}(x) > 0$ on $|x| \leq \log(a/2\pi)$, i.e. **(K3)** holds on the truncated domain. □

Immediate consequence. Since $w(a) > 0$ and **(K3)** holds for the explicit family, Remark 4.1 implies that $H_{\alpha,\beta}(x) > 0$ for all $x \in \mathbb{R}$.

4.2 Kernel positivity and scope

Theorem 4.2 establishes global pointwise positivity of the explicit kernel $H_{\alpha,\beta}(x)$ using only elementary inequalities and the variable-window/truncated representation from Section 3.

This ensures that any potential sign changes of the cosine transform associated with $H_{\alpha,\beta}$ cannot be attributed to pointwise negativity of the kernel itself. The next section shows that oscillatory effects may nevertheless arise from the *moving truncation boundary* and must therefore be analyzed separately via an explicit bulk–boundary decomposition.

5 Reduction to oscillatory boundary terms

In this section we analyze the cosine–transform functional associated with the kernel $H(x) = H_{\alpha,\beta}(x)$ and show that, despite the strict pointwise positivity established in Section 4, global nonnegativity of the transform does not follow automatically. We derive an explicit decomposition that isolates oscillatory boundary contributions generated by truncation, together with an interior remainder term.

5.1 Cosine–transform representation

By construction,

$$W(t) = \int_{-\infty}^{\infty} H(x)\, \cos(tx)\, dx, \qquad t \in \mathbb{R},$$

where H is even and strictly positive on $\mathbb{R}$.

Substituting the integral representation of $H(x)$ from Section 3 and applying Tonelli/Fubini (justified by absolute integrability) yields

$$W(t) = \int_{2\pi}^{\infty} w(a) \left(\int_{-\log(a/2\pi)}^{\log(a/2\pi)} P(a, x)\, e^{-a\cosh x} \cos(tx)\, dx \right) da.$$

Here the truncation of the inner integral arises directly from the moving lower bound $a \geq 2\pi e^{|x|}$ in the definition of $H(x)$.

5.2 Truncation and bulk–boundary decomposition

For each fixed $a \geq 2\pi$, define

$$L(a) = \log\Big(\frac{a}{2\pi}\Big), \qquad f_a(x) = P(a,x)\, e^{-a\cosh x}.$$

With this notation,

$$W(t) = \int_{2\pi}^{\infty} w(a) \left(\int_{-L(a)}^{L(a)} f_a(x)\cos(tx)\, dx \right) da.$$

Since f_a is even, this reduces to

$$W(t) = 2\int_{2\pi}^{\infty} w(a) \left(\int_{0}^{L(a)} f_a(x)\cos(tx)\, dx \right) da.$$

By Theorem 4.2 one has $f_a(x) > 0$ on $[0, L(a)]$ for every $a \geq 2\pi$. Nevertheless, the oscillatory integral $\int_0^{L(a)} f_a(x)\cos(tx)\, dx$ has no *a priori* sign. The next subsections therefore isolate explicitly the truncation–induced boundary contribution and separate it from an interior remainder whose sign (and possible dominance) requires additional harmonic comparison.

5.3 Finite-window oscillations and the boundary mechanism

Even when $f_a(x) > 0$ on $[0, L(a)]$ for each fixed a, the finite-window cosine integral

$$\int_0^{L(a)} f_a(x)\cos(tx)\, dx$$

has no *a priori* sign: its value depends on the frequency t and on the truncation length $L(a)$. The absence of sign control is therefore a *boundary* effect associated with the finite upper limit of integration. In the moving-truncation setting considered here, the cutoff at $x = L(a)$ generates an additional oscillatory contribution that is not detectable from pointwise kernel bounds alone. For this reason, to analyze potential sign reversals of $W(t)$ (and, in spectral/HAC contexts, the associated PSD concern), we next isolate the truncation-induced boundary term and separate it from an interior remainder by integration by parts.

5.4 Boundary isolation for $t \neq 0$

We now assume $t \neq 0$. Integrating by parts in x yields

$$\int_0^{L(a)} f_a(x)\cos(tx)\, dx = \frac{f_a(x)\sin(tx)}{t}\bigg|_0^{L(a)} - \frac{1}{t}\int_0^{L(a)} f_a'(x)\sin(tx)\, dx.$$

Since $\sin(0) = 0$, this reduces to

$$\int_0^{L(a)} f_a(x)\cos(tx)\, dx = \frac{f_a(L(a))\sin\big(tL(a)\big)}{t} - \frac{1}{t}\int_0^{L(a)} f_a'(x)\sin(tx)\, dx.$$

This identity separates a boundary contribution at $x = L(a)$ from an interior oscillatory remainder. No sign is asserted for either term individually; rather, the decomposition provides a transparent framework for tracking how truncation enters the sign of $W(t)$.

5.5 Structural interpretation

The bulk–boundary decomposition above is intentionally agnostic about the sign of the interior remainder. Its purpose is to make explicit the additional oscillatory component introduced by the moving truncation boundary. Concretely, the decomposition separates $W(t)$ into (i) a boundary contribution $B(t)$ coming solely from the cutoff at $x = L(a)$ and (ii) an interior remainder $R(t)$ involving oscillatory cancellations over a finite interval. Since neither term carries an intrinsic sign constraint, any global nonnegativity statement for W must control the boundary contribution relative to the interior remainder (equivalently, through an object governing their difference). This is the perspective adopted below when formulating a verifiable sufficient condition for global nonnegativity without presuming that either component is nonnegative on its own.

The case $t = 0$

For $t = 0$ the cosine–transform functional reduces to

$$W(0) = \int_{-\infty}^{\infty} H(x)\, dx.$$

Since $H(x) > 0$ for all $x \in \mathbb{R}$ and $H \in L^1(\mathbb{R})$, it follows immediately that

$$W(0) > 0.$$

5.6 Certified truncation bounds for numerical evaluation

To complement the analytic mechanism with rigorous evidence, we compute certified values of $W_{\alpha,\beta}(t)$ for reference parameters using validated numerics. The goal is not high-precision plotting, but a reproducible sign conclusion with explicit error control. We introduce truncated objects and bound the total discrepancy between $W_{\alpha,\beta}(t)$ and its finite computation by combining (i) deterministic tail bounds for truncation of the outer a–integral and the spatial x–integral, and (ii) an explicit quadrature error bound.

For $X > 0$ and $A > 0$ we introduce a truncated kernel and a truncated cosine transform. Let

$$H^A_{\alpha,\beta}(x) := \int_{2\pi e^{|x|}}^{A} w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}\, da, \tag{13}$$

with the convention that the integral is 0 whenever $A < 2\pi e^{|x|}$, and define

$$W^{A,X}_{\alpha,\beta}(t) := \int_{-X}^{X} H^A_{\alpha,\beta}(x) \cos(tx)\, dx. \tag{14}$$

The next lemma provides an explicit *a priori* bound for the truncation error $|W_{\alpha,\beta}(t) - W^{A,X}_{\alpha,\beta}(t)|$, uniformly in t.

Lemma 5.1 (Explicit truncation error bound)**.** *Assume $\alpha \in (0, 2\pi)$ and $\beta > 0$. Fix $X \geq 0$ and*

$$A \geq \max\{e^2,\, 2\pi e^X\}. \tag{15}$$

Set

$$C_{\alpha,\beta} := \frac{1}{8\pi}\,(2\pi)^{-3/2}\left(1 + \frac{\alpha}{2\pi} + \frac{\beta}{(2\pi)^2}\right). \tag{16}$$

Then for every $t \in \mathbb{R}$,

$$\left|W_{\alpha,\beta}(t) - W_{\alpha,\beta}^{A,X}(t)\right| \le E_A(X;A) + E_X(X), \tag{17}$$

where

$$E_A(X;A) := 2X\, C_{\alpha,\beta}\, e^{-A}\Big(A^4 + 4A^3 + 12A^2 + 24A + 24\Big), \tag{18}$$

$$E_X(X) := \frac{C_{\alpha,\beta}}{\pi}\, \exp\!\big(-\pi e^{2X}\big)\left(\frac{(4\pi)^4}{8}\, e^{X} + (4\pi)^3 + 6(4\pi)^2\, e^{-X} + 24(4\pi)\, e^{-2X} + 48\, e^{-3X}\right). \tag{19}$$

Proof. Since $|\cos(tx)| \le 1$, we have

$$|W_{\alpha,\beta}(t) - W_{\alpha,\beta}^{A,X}(t)| \le \int_{|x|>X} H_{\alpha,\beta}(x)\, dx + \int_{-X}^{X} \big(H_{\alpha,\beta}(x) - H_{\alpha,\beta}^{A}(x)\big)\, dx.$$

We bound each term using an integrable majorant.

First, for $a \ge 2\pi$ we have $\log(a/2\pi) \le \log a$. Under (15), we also have $\log a \le a^{1/2}$ for $a \ge e^2$. Hence

$$w(a) = \frac{1}{8\pi} \log\!\Big(\frac{a}{2\pi}\Big)\Big(\frac{a}{2\pi}\Big)^{3/2} \le \frac{1}{8\pi}(2\pi)^{-3/2}\, a^2.$$

Next, on the truncation domain $a \ge 2\pi e^{|x|}$ we use $\cosh x \le e^{|x|}$ to obtain

$$|P_{\alpha,\beta}(a,x)| = |a^2 + \beta - \alpha a \cosh x| \le a^2 + \beta + \alpha a e^{|x|} \le \Big(1 + \frac{\alpha}{2\pi}\Big)\, a^2 + \beta \le \Big(1 + \frac{\alpha}{2\pi} + \frac{\beta}{(2\pi)^2}\Big)\, a^2.$$

Combining the two bounds gives the pointwise majorant

$$w(a)\, |P_{\alpha,\beta}(a,x)|\, e^{-a\cosh x} \le C_{\alpha,\beta}\, a^4\, e^{-a\cosh x},$$

with $C_{\alpha,\beta}$ as in (16).

(1) Truncation in a on $[-X, X]$. For $|x| \le X$ one has $\cosh x \ge 1$, hence $e^{-a\cosh x} \le e^{-a}$, and

$$\int_{-X}^{X} \big(H_{\alpha,\beta}(x) - H_{\alpha,\beta}^{A}(x)\big)\, dx \le \int_{-X}^{X}\int_{A}^{\infty} C_{\alpha,\beta}\, a^4 e^{-a}\, da\, dx = 2X\, C_{\alpha,\beta} \int_{A}^{\infty} a^4 e^{-a}\, da.$$

A direct 4-fold integration by parts yields

$$\int_{A}^{\infty} a^4 e^{-a}\, da = e^{-A}\big(A^4 + 4A^3 + 12A^2 + 24A + 24\big),$$

which gives (18).

(2) Truncation in x on $|x| > X$. For $x \ge X$ we use $\cosh x \ge \frac{1}{2}e^x$ and the lower limit $a \ge 2\pi e^x$ to get

$$H_{\alpha,\beta}(x) \le \int_{2\pi e^x}^{\infty} C_{\alpha,\beta}\, a^4\, e^{-(e^x/2)a}\, da.$$

Let $\lambda = e^x/2$ and $a_0 = 2\pi e^x = 4\pi\lambda$. A 4-fold integration by parts gives

$$\int_{a_0}^{\infty} a^4 e^{-\lambda a}\, da = e^{-\lambda a_0}\left(\frac{a_0^4}{\lambda} + \frac{4a_0^3}{\lambda^2} + \frac{12a_0^2}{\lambda^3} + \frac{24a_0}{\lambda^4} + \frac{24}{\lambda^5}\right).$$

Since $\lambda a_0 = \pi e^{2x}$ and $a_0 = 4\pi\lambda$, this becomes

$$\int_{2\pi e^x}^{\infty} a^4 e^{-(e^x/2)a}\, da = \exp\!\big(-\pi e^{2x}\big)\Big(\frac{(4\pi)^4}{8}\, e^{3x} + (4\pi)^3 e^{2x} + 6(4\pi)^2 e^x + 24(4\pi) + 48e^{-x}\Big).$$

Therefore,

$$\int_X^{\infty} H_{\alpha,\beta}(x)\, dx \le C_{\alpha,\beta} \int_X^{\infty} \exp\!\big(-\pi e^{2x}\big)\Big(\frac{(4\pi)^4}{8}\, e^{3x} + (4\pi)^3 e^{2x} + 6(4\pi)^2 e^x + 24(4\pi) + 48e^{-x}\Big)\, dx.$$

With the substitution $u = e^{2x}$, $dx = \frac{du}{2u}$, each term becomes an integral of the form $\int_{e^{2X}}^{\infty} u^p e^{-\pi u}\, du$. Since $u \mapsto u^p e^{-\pi u}$ is decreasing on $[e^{2X}, \infty)$, one has the bound $\int_{e^{2X}}^{\infty} u^p e^{-\pi u}\, du \le \frac{1}{\pi} e^{-\pi e^{2X}} (e^{2X})^p$. Collecting terms and using symmetry ($\int_{|x|>X} = 2\int_X^{\infty}$) yields (19). □

Corollary 5.2 (Numerical scale for the reference kernel)**.** *For $(\alpha, \beta) = (6, 9)$, choosing for instance $X = 3$ and $A = 200$ (which satisfies $A \ge 2\pi e^X$) gives*

$$E_A(3; 200) + E_X(3) \;<\; 10^{-70}.$$

Hence any numerically observed negative value of size $\gtrsim 10^{-9}$ cannot be an artifact of domain truncation.

Lemma 5.3 (Validated truncation–quadrature bound)**.** *Fix $X > 0$, $A > 2\pi e^X$ and an integer $N \ge 1$, and set $h = X/N$. Let*

$$H_A(x) := \int_{2\pi e^{|x|}}^{A} w(a)\, P_{\alpha,\beta}(a, x)\, e^{-a\cosh x}\, da, \qquad W_{\alpha,\beta}(t) = \int_{\mathbb{R}} H_{\alpha,\beta}(x) \cos(tx)\, dx.$$

Define the (outer) trapezoidal approximation on $[0, X]$ by

$$\widetilde{W}_{A,X,N}(t) := 2h\Big(\tfrac{1}{2} H_A(0) + \sum_{j=1}^{N-1} H_A(jh)\cos(tjh) + \tfrac{1}{2} H_A(X)\cos(tX)\Big).$$

Then for every $t \in \mathbb{R}$,

$$\big|W_{\alpha,\beta}(t) - \widetilde{W}_{A,X,N}(t)\big| \le E_X(X) + E_A(X; A) + E_{\mathrm{trap}}(t; A, X, N),$$

where

$$E_A(X; A) := 2X \int_A^{\infty} w(a)\big(a^2 + \beta + \alpha a \cosh X\big) e^{-a}\, da,$$

$$E_X(X) := 2\int_X^{\infty} \int_{2\pi e^x}^{\infty} w(a)\big(a^2 + \beta + \alpha a \cosh x\big) e^{-a\cosh x}\, da\, dx,$$

and the trapezoidal error satisfies the explicit bound

$$E_{\mathrm{trap}}(t; A, X, N) \le \frac{Xh^2}{6}\, C_2(t), \qquad C_2(t) := M_2 + 2|t| M_1 + t^2 M_0,$$

with

$$M_0 := \int_{2\pi}^{A} w(a)\big(a^2 + \beta + \alpha a \cosh X\big) e^{-a}\, da,$$

$$M_1 := \max_{u \in [0,X]} \rho_{\alpha,\beta}(u) + \int_{2\pi}^{A} w(a)\, a\big(a^2 + \beta + \alpha a \cosh X + \alpha\big) e^{-a}\, da,$$

$$M_2 := B_\partial(X) + \int_{2\pi}^{A} w(a)\Big((a+a^2)\big(a^2+\beta+\alpha a\cosh X+\alpha\big)+\alpha a^2\Big)e^{-a}\,da,$$

where $\rho_{\alpha,\beta}$ is the boundary density in (25)*, and $B_\partial(X)$ is the explicit boundary contribution coming from Leibniz differentiation of the moving lower limit $a = 2\pi e^x$ (for $x \in [0, X]$), namely*

$$B_\partial(X) := \max_{u\in[0,X]}\Big(\rho_{\alpha,\beta}(u) + 2(2\pi e^u)\big|F_x(2\pi e^u, u)\big| + (2\pi e^u)^2\big|F_a(2\pi e^u, u)\big|\Big),$$

with $F(a, x) := w(a)P_{\alpha,\beta}(a, x)e^{-a\cosh x}$.

5.7 Explicit boundary density and high–frequency decay

We keep the notation introduced above:

$$L(a) = \log\Big(\frac{a}{2\pi}\Big), \qquad f_a(x) = P(a, x)e^{-a\cosh x}.$$

For $t \neq 0$, the integration–by–parts identity yields the decomposition

$$W(t) = \frac{2}{t}\,B(t) - \frac{2}{t}\,R(t), \tag{20}$$

where

$$B(t) = \int_{2\pi}^{\infty} w(a)\, f_a\big(L(a)\big)\,\sin\big(tL(a)\big)\,da, \tag{21}$$

and

$$R(t) = \int_{2\pi}^{\infty} w(a)\int_0^{L(a)} f_a'(x)\,\sin(tx)\,dx\,da. \tag{22}$$

5.7.1 Closed form of the boundary density

The boundary term (21) can be written as a sine transform after the change of variables $u = L(a) = \log(a/2\pi)$, namely

$$B(t) = \int_0^{\infty} \rho(u)\,\sin(tu)\,du, \tag{23}$$

with density

$$\rho(u) = w(2\pi e^u)\, f_{2\pi e^u}(u)\,(2\pi e^u), \qquad u \geq 0. \tag{24}$$

In the present setting the density admits a closed form. Indeed, since

$$w(2\pi e^u) = \frac{1}{8\pi}\,u\,e^{3u/2}, \qquad (2\pi e^u)\cosh u = \pi\big(e^{2u}+1\big),$$

we obtain for the two–parameter family $P_{\alpha,\beta}(a, x) = a^2 + \beta - \alpha a\cosh x$ the identity

$$\rho_{\alpha,\beta}(u) = \frac{u}{4}\,e^{5u/2}\,\Big(\big(4\pi^2-\alpha\pi\big)e^{2u} + (\beta-\alpha\pi)\Big)\,\exp\big(-\pi(e^{2u}+1)\big), \qquad u \geq 0. \tag{25}$$

In particular, for the reference parameters $(\alpha, \beta) = (6, 9)$ we write $\rho = \rho_{6,9}$.

Lemma 5.4 (Regularity and super–exponential decay)**.** *Fix $(\alpha, \beta) \in (0, 2\pi) \times (0, \infty)$. The density $\rho_{\alpha,\beta}$ defined in* (25) *satisfies:*

1. *$\rho_{\alpha,\beta} \in C^\infty([0, \infty))$, $\rho_{\alpha,\beta}(0) = 0$, and $\rho_{\alpha,\beta}(u) \to 0$ as $u \to \infty$.*

2. *For every $m \geq 0$, the derivative $\rho_{\alpha,\beta}^{(m)}$ belongs to $L^1(0,\infty)$.*

Proof. Both items follow from the explicit form (25). The density is a product of a smooth factor of at most exponential growth (polynomials in u and e^u) and the super–exponentially decaying term $\exp(-\pi e^{2u})$. Differentiation preserves this structure and yields bounds of the form

$$|\rho_{\alpha,\beta}^{(m)}(u)| \leq C_m \exp\bigl(-\tfrac{\pi}{2}e^{2u}\bigr), \qquad u \geq 0,$$

for suitable constants $C_m > 0$, implying $\rho_{\alpha,\beta}^{(m)} \in L^1(0,\infty)$ for all m. Moreover, $\rho_{\alpha,\beta}(0) = 0$ due to the prefactor u in (25), and decay at infinity is immediate. □

Proposition 5.5 (Boundary asymptotics)**.** *Let $\rho \in C^\infty([0,\infty))$ satisfy $\rho^{(m)} \in L^1(0,\infty)$ for all $m \geq 0$ and $\rho(u) \to 0$ as $u \to \infty$. Define*

$$B(t) := \int_0^\infty \rho(u)\, \sin(tu)\, du.$$

Then for every integer $M \geq 1$ and every $t \neq 0$,

$$B(t) = \sum_{n=1}^{M} (-1)^n \frac{\rho^{(2n)}(0)}{t^{2n+1}} + \mathcal{O}\Bigl(\frac{1}{|t|^{2M+3}}\Bigr) \qquad (|t| \to \infty), \tag{26}$$

where the implied constant depends only on $\|\rho^{(2M+2)}\|_{L^1(0,\infty)}$. In particular, if $\rho^{(2)}(0) \neq 0$ then $B(t) \sim -\rho^{(2)}(0)/t^3$ as $|t| \to \infty$.

Proof. Repeated integration by parts in the oscillatory integral yields, for $t \neq 0$,

$$B(t) = -\frac{\rho^{(2)}(0)}{t^3} + \frac{1}{t^3}\int_0^\infty \rho^{(3)}(u)\cos(tu)\, du,$$

and iterating the same argument M times gives (26), with the remainder bounded by $|t|^{-(2M+3)}\|\rho^{(2M+2)}\|_{L^1(0,\infty)}$. □

Corollary 5.6 (Explicit leading coefficient)**.** *For the explicit density $\rho_{\alpha,\beta}$ in (25) one has*

$$\rho''_{\alpha,\beta}(0) = e^{-2\pi}\Bigl[\alpha\Bigl(2\pi^2 - \frac{7\pi}{2}\Bigr) + \beta\Bigl(\frac{5}{4} - \pi\Bigr) + \pi^2(9 - 4\pi)\Bigr]. \tag{27}$$

In particular, for $(\alpha,\beta) = (6,9)$ one obtains $\rho''_{6,9}(0) > 0$, hence

$$B_{6,9}(t) \sim -\frac{\rho''_{6,9}(0)}{t^3} < 0 \qquad (|t| \to \infty).$$

Proof. Expand (25) in a Taylor series at $u = 0$ up to order u^2 and identify the u^2–coefficient. □

Lemma 5.7 (Exact cancellation identity)**.** *For $x \geq 0$ define*

$$H(x) = \int_{2\pi e^x}^\infty w(a)\, P_{\alpha,\beta}(a,x)\, e^{-a\cosh x}\, da, \qquad \rho(x) = (2\pi e^x)\, w(2\pi e^x)\, P_{\alpha,\beta}(2\pi e^x, x)\, e^{-(2\pi e^x)\cosh x}.$$

Then, for every $t \in \mathbb{R}$,

$$B(t) - R(t) = -\int_0^\infty H'(x)\, \sin(tx)\, dx, \tag{28}$$

and consequently, for $t \neq 0$,

$$W(t) = \frac{2}{t}\bigl(B(t) - R(t)\bigr) = -\frac{2}{t}\int_0^\infty H'(x)\sin(tx)\, dx,$$

which coincides with the one–step integration–by–parts identity for $W(t) = 2\int_0^\infty H(x)\cos(tx)\, dx$.

Proof. For $x \geq 0$, differentiate $H(x) = \int_{2\pi e^x}^{\infty} F(a,x)\,da$ with $F(a,x) = w(a)P_{\alpha,\beta}(a,x)e^{-a\cosh x}$. Leibniz' rule gives

$$H'(x) = -(2\pi e^x)F(2\pi e^x, x) + \int_{2\pi e^x}^{\infty} \partial_x F(a,x)\,da = -\rho(x) + \int_{2\pi e^x}^{\infty} w(a)\, f_a'(x)\,da.$$

Denoting $r(x) := \int_{2\pi e^x}^{\infty} w(a)\, f_a'(x)\,da$, we have $r(x) = H'(x) + \rho(x)$ and hence

$$R(t) = \int_0^{\infty} r(x)\sin(tx)\,dx = \int_0^{\infty} \rho(x)\sin(tx)\,dx + \int_0^{\infty} H'(x)\sin(tx)\,dx = B(t) + \int_0^{\infty} H'(x)\sin(tx)\,dx,$$

which implies (28). □

Corollary 5.8 (Beyond–all–orders decay of W)**.** *Assume that H is even and $\partial_x^m H \in L^1(0,\infty)$ for all $m \geq 0$. Then for every $N \geq 1$ there exists $C_N > 0$ such that*

$$|W(t)| \leq \frac{C_N}{(1+|t|)^N}, \qquad t \in \mathbb{R}.$$

Proof. For $t \neq 0$, write $W(t) = 2\int_0^{\infty} H(x)\cos(tx)\,dx$ and integrate by parts N times. Boundary terms at $x = 0$ vanish by evenness (odd derivatives at 0), and boundary terms at $x = \infty$ vanish by decay. The remainder is bounded by $\|\partial_x^N H\|_{L^1(0,\infty)}/|t|^N$, and a standard adjustment yields the stated $(1+|t|)^{-N}$ bound for all $t \in \mathbb{R}$. □

5.7.2 Schwartz decay of the full transform

We next record a corresponding rapid-decay bound for $W_{\alpha,\beta}(t)$ obtained directly from the variable-window representation and the envelope bounds established in Section 3.

Proposition 5.9 (Rapid decay under variable-window envelope conditions)**.** *Assume that H admits a representation of the form* (3) *and that the bookkeeping conditions **(K1)**–**(K2)** hold for the associated family $\{f_a\}$ and weight w. Let W be the cosine transform of H. Then for every integer $N \geq 1$ there exists $C_N > 0$ such that*

$$|W(t)| \leq \frac{C_N}{(1+|t|)^N}, \qquad t \in \mathbb{R}.$$

Proof. Let $W(t) = \int_{\mathbb{R}} H(x)\cos(tx)\,dx$. By Lemma 3.2,

$$W(t) = 2\int_C^{\infty} w(a)\left(\int_0^{L(a)} f_a(x)\cos(tx)\,dx\right)da, \qquad L(a) = \log(a/C).$$

Fix an integer $N \geq 1$ and assume $t \neq 0$. For each fixed $a \geq C$, integrate by parts N times in x:

$$\int_0^{L(a)} f_a(x)\cos(tx)\,dx = \sum_{k=0}^{N-1} \frac{E_k(a,t)}{t^{k+1}} + \frac{1}{t^N}\int_0^{L(a)} \partial_x^N f_a(x)\,\Phi_N(tx)\,dx,$$

where Φ_N is either sin or cos and

$$|E_k(a,t)| \leq |\partial_x^k f_a(0)| + |\partial_x^k f_a(L(a))|.$$

Using **(K1)** on $x \in [0, L(a)]$ and $\cosh x \geq 1$, we obtain

$$|\partial_x^m f_a(x)| \leq c_m(1+a)^{p_m}e^{-a} \qquad (0 \leq x \leq L(a)).$$

Hence

$$\int_0^{L(a)} |\partial_x^N f_a(x)|\, dx \le c_N(1+a)^{p_N} L(a) e^{-a}, \qquad |\partial_x^k f_a(0)| \le c_k(1+a)^{p_k} e^{-a}.$$

At the moving endpoint $x = L(a)$, we use

$$\cosh(L(a)) = \frac{1}{2}\Big(\frac{a}{C} + \frac{C}{a}\Big) \ge \frac{a}{2C},$$

so $e^{-a\cosh(L(a))} \le e^{-a^2/(2C)}$, which implies the stronger bound $|\partial_x^k f_a(L(a))| \le \widetilde{c}_k(1+a)^{\widetilde{p}_k} e^{-a^2/(2C)}$.

Combining these estimates yields

$$\left| \int_0^{L(a)} f_a(x)\cos(tx)\, dx \right| \le \frac{A_N(a)}{|t|^N},$$

where $A_N(a)$ is bounded by a finite linear combination of terms of the form $(1+a)^q L(a) e^{-a}$ and $(1+a)^q e^{-a^2/(2C)}$ (with coefficients depending on N). Multiplying by $w(a)$ and integrating over $a \in [C, \infty)$, **(K2)** (and $L(a) \ll (1+a)$) yields

$$|W(t)| \le \frac{C_N}{|t|^N} \qquad (t \ne 0).$$

Since W is continuous and bounded near $t = 0$, this extends to $|W(t)| \le C_N(1+|t|)^{-N}$ for all $t \in \mathbb{R}$. □

Remark 5.10 (Interpretation). Proposition 5.9 clarifies why numerically observed sign changes for the reference kernel can occur at extremely small magnitudes. While the boundary contribution $B(t)$ admits an explicit algebraic high-frequency expansion (Proposition 5.5), the exact identity (28) forces cancellation between the boundary and interior contributions in the combination $B(t) - R(t)$ that determines $W(t)$. Consequently, $W(t)$ decays rapidly in $|t|$ and any negative excursions—when present—occur on a small numerical scale.

6 A sufficient criterion for global nonnegativity

This section isolates a verifiable sufficient criterion for global nonnegativity of the cosine transform

$$W(t) = \int_{\mathbb{R}} H(x)\cos(tx)\, dx$$

for kernels with spatially dependent truncation windows. The key point is that the bulk–boundary analysis from the previous section can be rewritten in terms of a single *defect density* whose sine transform controls the sign of W.

6.1 Reduction to a sine transform

We keep the notation introduced above. For $a \ge 2\pi$ let

$$L(a) = \log\Big(\frac{a}{2\pi}\Big), \qquad f_a(x) = P(a,x)e^{-a\cosh x},$$

and recall the decomposition (for $t \ne 0$)

$$W(t) = \frac{2}{t}B(t) - \frac{2}{t}R(t),$$

where

$$B(t) = \int_{2\pi}^{\infty} w(a)\, f_a(L(a))\, \sin\bigl(tL(a)\bigr)\, da, \qquad R(t) = \int_{2\pi}^{\infty} w(a) \int_0^{L(a)} f_a'(x)\, \sin(tx)\, dx\, da.$$

After the change of variables $u = L(a)$, the boundary term becomes a sine transform

$$B(t) = \int_0^{\infty} \rho(u) \sin(tu)\, du, \qquad \rho(u) = (2\pi e^u)\, w(2\pi e^u)\, f_{2\pi e^u}(u).$$

6.2 Laplace mixtures and positive sine transforms

Definition 6.1 (Completely monotone Laplace mixtures)**.** A function $g : [0, \infty) \to \mathbb{R}$ is said to be a Laplace mixture if there exists a finite nonnegative Borel measure μ on $(0, \infty)$ such that

$$g(u) = \int_0^{\infty} e^{-su}\, d\mu(s), \qquad u \geq 0.$$

Equivalently, g is completely monotone on $(0, \infty)$.

Lemma 6.2 (Positivity of a sine transform)**.** *Let g be a Laplace mixture in the sense of Definition 6.1, and define*

$$\eta(u) = u\, g(u), \qquad u \geq 0.$$

Then for every $t \geq 0$,

$$\int_0^{\infty} \eta(u) \sin(tu)\, du \;\geq\; 0.$$

More precisely,

$$\int_0^{\infty} \eta(u) \sin(tu)\, du = \int_0^{\infty} \frac{2st}{(s^2 + t^2)^2}\, d\mu(s), \qquad t \geq 0,$$

and the right-hand side is nonnegative.

Proof. Fix $t \geq 0$. Since ue^{-su} is integrable on $(0, \infty)$ for each $s > 0$ and μ is finite, Tonelli's theorem gives

$$\int_0^{\infty} \eta(u) \sin(tu)\, du = \int_0^{\infty} \left(\int_0^{\infty} ue^{-su} \sin(tu)\, du \right) d\mu(s).$$

A direct computation yields, for $s > 0$,

$$\int_0^{\infty} ue^{-su} \sin(tu)\, du = \frac{2st}{(s^2 + t^2)^2} \geq 0,$$

which proves the claim. □

6.3 A verifiable criterion via the defect density

The cancellation identity in Lemma 5.7 shows that the relevant object for sign control is the *difference* between boundary and interior contributions, which can be expressed in terms of the derivative of the kernel.

Proposition 6.3 (Defect–density criterion (verifiable form))**.** *Assume that H is even, $H \in C^1(\mathbb{R})$, $H \in L^1(\mathbb{R})$, and $H' \in L^1(0,\infty)$. Define the defect density*

$$\kappa(u) := -H'(u), \qquad u \geq 0.$$

Then for every $t > 0$,

$$W(t) = \frac{2}{t} \int_0^\infty \kappa(u) \sin(tu)\, du, \tag{29}$$

and $W(0) = \int_{\mathbb{R}} H(x)\, dx > 0$. Consequently, the following are equivalent:

(i) *$W(t) \geq 0$ for all $t \in \mathbb{R}$;*

(ii) $\displaystyle\int_0^\infty \kappa(u) \sin(tu)\, du \geq 0$ *for all $t \geq 0$.*

A concrete sufficient condition for (ii) *is that κ admits the Laplace–mixture form*

$$\kappa(u) = u \int_0^\infty e^{-su}\, d\mu(s) \qquad (u \geq 0) \tag{30}$$

for some finite nonnegative Borel measure μ on $(0,\infty)$. In that case, for every $t \geq 0$,

$$\int_0^\infty \kappa(u) \sin(tu)\, du = \int_0^\infty \frac{2st}{(s^2+t^2)^2}\, d\mu(s) \geq 0,$$

and therefore $W(t) \geq 0$ for all $t \in \mathbb{R}$.

Proof. For $t > 0$, write $W(t) = 2\int_0^\infty H(x)\cos(tx)\, dx$ and integrate by parts once to obtain

$$W(t) = -\frac{2}{t} \int_0^\infty H'(x) \sin(tx)\, dx = \frac{2}{t} \int_0^\infty \kappa(x) \sin(tx)\, dx,$$

which is (29). Since $2/t > 0$ for $t > 0$, the equivalence between (i) and (ii) follows immediately, and $W(0) > 0$ holds because $H > 0$ and $H \in L^1(\mathbb{R})$ in our setting. If (30) holds, Lemma 6.2 applies to $\kappa(u) = u\, g(u)$ with $g(u) = \int_0^\infty e^{-su}\, d\mu(s)$ and yields nonnegativity for all $t \geq 0$. Finally, since $H' \in L^1(0,\infty)$, the function H has a finite limit as $x \to +\infty$; combined with $H \in L^1(\mathbb{R})$ this limit must be 0, so the boundary term at $+\infty$ vanishes in the integration-by-parts step. □

Remark 6.4 (A crude comparison bound). Starting from the bulk–boundary decomposition $W(t) = \frac{2}{t}(B(t) - R(t))$ and applying the triangle inequality to the interior term gives the sufficient condition

$$B(t) \;\geq\; \int_{2\pi}^\infty w(a) \int_0^{L(a)} |f_a'(x)|\, |\sin(tx)|\, dx\, da \quad (t \geq 0),$$

which implies $W(t) \geq 0$. This bound is easy to state but typically conservative; Proposition 6.3 identifies the sharper object controlling the sign, namely the sine transform of $\kappa = -H'$.

6.4 A concrete positive test case

We give a simple explicit instance where the sufficient mechanism in Proposition 6.3 is satisfied, illustrating that the criterion is nonvacuous.

Proposition 6.5 (Explicit Laplace–mixture defect density)**.** *Fix constants $c_1, c_2 > 0$ and $s_1, s_2 > 0$, and define the discrete measure*

$$\mu := c_1\,\delta_{s_1} + c_2\,\delta_{s_2}.$$

Let

$$\kappa(u) := u\int_0^\infty e^{-su}\,d\mu(s) = u\Big(c_1 e^{-s_1 u} + c_2 e^{-s_2 u}\Big), \qquad u \geq 0.$$

Then for every $t \geq 0$,

$$\int_0^\infty \kappa(u)\sin(tu)\,du = 2t\left(\frac{c_1 s_1}{(s_1^2+t^2)^2} + \frac{c_2 s_2}{(s_2^2+t^2)^2}\right) \geq 0,$$

and therefore any even kernel H with $-(H)' = \kappa$ satisfies $W(t) \geq 0$ for all $t \in \mathbb{R}$.

Proof. This is Lemma 6.2 specialized to the two–atom measure $\mu = c_1\delta_{s_1} + c_2\delta_{s_2}$. □

7 Monte Carlo Evidence

7.1 Purpose and link to the mechanism

The theory sections isolate a boundary-induced mechanism by which *moving truncations* can generate negative spectral estimates and non–PSD covariance estimators, even when the underlying kernel is strictly positive. The key point is that a truncation with a *state-dependent boundary* produces an explicit oscillatory boundary contribution (a sine/Dirichlet-type term) plus an interior remainder; their interaction can overturn naive positivity intuition at very small magnitude scales. This section provides Monte Carlo evidence that (i) such nonnegativity/PSD failures occur with nontrivial frequency in finite samples under empirically relevant nonstationarities, and (ii) they can matter for inference through coverage and test-size distortions when HAC standard errors are computed from non-admissible (negative / non–PSD) long-run covariance estimates.

Our simulations are designed to fill three applied gaps: (i) *frequency and magnitude* of negativity / non–PSD under finite T; (ii) *impact on inference* (coverage and size/power of HAC-based procedures); (iii) a *systematic parameter map* for sign changes in the analytic family $H_{\alpha,\beta}$ (Step 5).

7.2 Data-generating processes

We consider two DGPs, each chosen to stress the types of adaptivity used in practice (variance changes and regime persistence), while keeping the regression design transparent.

DGP1 (AR(1) with conditional heteroskedasticity). Let $\{z_t\}$ be i.i.d. $\mathcal{N}(0,1)$. Generate

$$u_t = \phi u_{t-1} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \tag{31}$$

$$\sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2, \tag{32}$$

with $(\phi, \omega, a, b) = (0.6,\ 0.05,\ 0.10,\ 0.85)$ so that $a + b < 1$.

DGP2 (Markov-switching persistence and variance). Let $s_t \in \{0,1\}$ be a two-state Markov chain with transition matrix

$$\mathbb{P}(s_t = 1 \mid s_{t-1} = 1) = p_{11}, \qquad \mathbb{P}(s_t = 0 \mid s_{t-1} = 0) = p_{00},$$

with $(p_{00}, p_{11}) = (0.97, 0.90)$. Generate

$$u_t = \phi_{s_t} u_{t-1} + \sigma_{s_t} z_t, \qquad z_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \tag{33}$$

$$(\phi_0, \sigma_0) = (0.2, 1.0), \qquad (\phi_1, \sigma_1) = (0.8, 3.0). \tag{34}$$

7.3 Targets: spectra and long-run covariances

Spectral targets. For a scalar series $\{v_t\}$ (e.g. u_t), define

$$f_v(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_v(k) e^{-i\omega k}, \qquad \gamma_v(k) = \mathrm{Cov}(v_t, v_{t-k}).$$

We estimate $f_v(\omega)$ on a grid $\omega_j = \pi j/J$, $j = 0, 1, \ldots, J$, with $J = 256$ in the reported runs.

HAC targets for regression inference. Consider

$$y_t = x_t^\top \beta + u_t, \qquad t = 1, \ldots, T,$$

with OLS $\widehat{\beta}$ and score

$$s_t = x_t \widehat{u}_t, \qquad \widehat{u}_t = y_t - x_t^\top \widehat{\beta}.$$

Let $\Omega = \sum_{k=-\infty}^{\infty} \Gamma(k)$ be the long-run covariance of $\{s_t\}$, where $\Gamma(k) = \mathbb{E}[s_t s_{t-k}^\top]$. A generic HAC estimator $\widehat{\Omega}$ induces

$$\widehat{\mathrm{Var}}(\widehat{\beta}) = \frac{1}{T} (X^\top X/T)^{-1} \widehat{\Omega} (X^\top X/T)^{-1}.$$

We study both: (i) a simple regression with $x_t = (1, x_{1t})^\top$, and (ii) a multivariate regression with $x_t = (1, x_{1t}, x_{2t})^\top$.

Regressor designs. We keep regressors exogenous and mildly persistent:

$$x_{jt} = \rho_x x_{j,t-1} + \eta_{jt}, \qquad \eta_{jt} \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \qquad \rho_x = 0.3,$$

independent of $\{u_t\}$. We set $\beta = (1, 0, 0)^\top$ for size/coverage experiments and $\beta = (1, \delta/\sqrt{T}, 0)^\top$ for local-power experiments.

7.4 Compared estimators and PSD repair

All estimators below are applied to the score sequence $\{s_t\}$ (scalar or vector).

E1: Newey–West / Bartlett with fixed bandwidth (baseline). Fix $m_0 = \lfloor c_0 T^{1/3} \rfloor$ with $c_0 = 1.5$ in the reported runs. Define $K_{\mathrm{B}}(x) = (1 - |x|)\mathbf{1}\{|x| \le 1\}$ and set

$$\widehat{\Omega}_{\mathrm{E1}} = \sum_{k=-m_0}^{m_0} K_{\mathrm{B}}\left(\frac{k}{m_0 + 1}\right) \widehat{\Gamma}(k).$$

This Bartlett/Newey–West construction is PSD-preserving in finite samples under the standard spectral-window interpretation Bartlett, 1950; Newey and West, 1987.

E2: Standard data-dependent bandwidth (Andrews-style global adaptivity). Use a quadratic-spectral kernel with an automatic plug-in bandwidth $\widehat{b}_{\mathrm{A}}$ (AR(1) plug-in, aggregated across score components) and set

$$\widehat{\Omega}_{\mathrm{E2}} = \sum_{k=-\lfloor \widehat{b}_{\mathrm{A}} \rfloor}^{\lfloor \widehat{b}_{\mathrm{A}} \rfloor} K_{\mathrm{QS}}\left(\frac{k}{\widehat{b}_{\mathrm{A}}}\right) \widehat{\Gamma}(k),$$

following the standard automatic-bandwidth HAC construction Andrews, 1991.

E0: Fixed *hard-cutoff* truncation (rectangular control).

$$\widehat{\Omega}_{\mathrm{E0}} := \sum_{k=-m_0}^{m_0} \widehat{\Gamma}(k), \qquad m_0 = \lfloor c_0 T^{1/3} \rfloor, \quad c_0 = 1.5.$$

This estimator is not PSD-preserving in finite samples and serves as a control for hard-cutoff effects without state dependence.

E3h: State-dependent *hard-cutoff* moving truncation (mechanism-targeted). Let $\widehat{u}_t$ be OLS residuals and define a robust volatility state

$$\widehat{\nu}_t := \log(\epsilon + \widehat{u}_t^2), \qquad \widetilde{\nu}_t := \frac{\widehat{\nu}_t - \mathrm{median}(\widehat{\nu}_{1:T})}{\mathrm{MAD}(\widehat{\nu}_{1:T})}, \qquad \epsilon = 10^{-8},$$

and $g(z) = (1 + e^{-z})^{-1}$. Define the bandwidth path

$$m_t := m_{\min} + \lfloor (m_{\max} - m_{\min})\, g(\kappa\, \widetilde{\nu}_t) \rfloor, \qquad (m_{\min}, m_{\max}, \kappa) = (2,\ \lfloor 2T^{1/3} \rfloor,\ 1.0).$$

Then define the moving-truncation long-run covariance estimator

$$\widehat{\Omega}_{\mathrm{E3h}} := \frac{1}{T} \sum_{t=1}^{T} \sum_{k=-m_t}^{m_t} s_t s_{t-k}^{\top}, \qquad s_{t-k} = 0 \text{ if } t-k \notin \{1, \ldots, T\}.$$

E3s: State-dependent tapered moving bandwidth (smooth analogue). Using the same $\{m_t\}$ as E3h, replace the hard cutoff weights by a Bartlett taper:

$$\widehat{\Omega}_{\mathrm{E3s}} := \frac{1}{T} \sum_{t=1}^{T} \sum_{k=-m_t}^{m_t} K_{\mathrm{B}}\left(\frac{k}{m_t + 1}\right) s_t s_{t-k}^{\top}, \qquad s_{t-k} = 0 \text{ if } t-k \notin \{1, \ldots, T\}.$$

E3s removes the discontinuity at $k = \pm m_t$ while preserving state dependence.

E4 (repair, reported): PSD projection by eigenvalue clipping. When $\widehat{\Omega}$ is not PSD, define

$$\widehat{\Omega}^{+} := Q\, \mathrm{diag}(\max\{\widehat{\lambda}_j, 0\})\, Q^{\top}, \quad \text{where } \widehat{\Omega} = Q\, \mathrm{diag}(\widehat{\lambda}_j)\, Q^{\top}.$$

We report the effect of this repair on size and coverage (columns Size5_PSD and Coverage95_PSD). (Shrinkage is a distinct regularization strategy; see, e.g., Ledoit and Wolf, 2004.)

7.5 A discrete bulk–boundary identity (summation by parts)

Lemma 7.1 (Discrete bulk–boundary decomposition via Abel transform)**.** *Let $\{\gamma(k)\}_{k\geq 0}$ be any real sequence and let $\omega \in (0,\pi)$. Define*

$$S_m(\omega) := \sum_{k=0}^{m} \gamma(k)\cos(\omega k).$$

Then

$$S_m(\omega) = \frac{\sin\big((m+\frac{1}{2})\omega\big)}{2\sin(\omega/2)}\,\gamma(m) \;+\; \sum_{k=0}^{m-1}\left(\frac{\sin\big((k+\frac{1}{2})\omega\big)}{2\sin(\omega/2)}\right)\big(\gamma(k)-\gamma(k+1)\big).$$

Remark 7.2 (Standard tool; role here is interpretive)*.* Lemma 7.1 is a standard Abel (summation-by-parts) identity combined with the Dirichlet-kernel representation Hardy, 1949; Katznelson, 2004. Its purpose here is interpretive: it makes the *boundary factor* at lag m explicit, so that when $m = m_t$ is state-dependent the discrete sum exhibits a literal *moving boundary*, mirroring the continuous bulk–boundary structure.

Interpretation. When $m = m_t$ depends on a state, the boundary factor at lag m_t becomes a moving boundary. The hard-cutoff construction E3h is designed to make this effect visible in finite samples through a discontinuity at $k = \pm m_t$, while E0 isolates hard-cutoff effects without state dependence. The tapered variant E3s holds the same $\{m_t\}$ fixed but replaces the discontinuous cutoff by a Bartlett taper; in the reported designs, no admissibility failures were observed for E3s, and its behavior is closer to the PSD-preserving baselines (E1–E2). This pattern is consistent with the idea that discontinuities induced by hard cutoffs (especially when state-dependent) can amplify boundary-driven ringing under the chosen state rule.

7.6 Monte Carlo metrics and numerical integrity

Spectral diagnostics (univariate).

$$\mathbb{I}\{\min_{\omega\in\mathcal{G}} \widehat{f}(\omega) < 0\}.$$

PSD diagnostics (multivariate).

$$\mathbb{I}\{\lambda_{\min}(\widehat{\Omega}) < 0\}.$$

Numerical integrity (required). All PSD diagnostics are computed after explicit symmetrization:

$$\widehat{\Omega} \leftarrow \tfrac{1}{2}(\widehat{\Omega} + \widehat{\Omega}^\top),$$

and we report the symmetry defect $\|\widehat{\Omega} - \widehat{\Omega}^\top\|_{\max}$ (reported as Asym~ 0 in all runs).

Inference diagnostics (coverage, size, power). For coefficient β_1:

- **95% CI coverage:** $\mathbb{P}\{\beta_1 \in [\widehat{\beta}_1 \pm 1.96\sqrt{\widehat{V}_{11}}]\}$.
- **Size at 5%:** rejection frequency under $H_0 : \beta_1 = 0$.
- **Power (local):** rejection frequency under $\beta_1 = \delta/\sqrt{T}$ with $\delta = 1$ (reported in Figure 4).

7.7 Monte Carlo design: sizes, replications, and tuning

We use $T \in \{250, 500, 1000\}$ and $R = 2000$ replications for $T \in \{250, 500\}$ and $R = 1000$ for $T = 1000$. The spectral grid is $J = 256$ on $[0, \pi]$. For E1 and E0 we set $c_0 = 1.5$; E2 uses the standard QS plug-in bandwidth; E3h uses $(m_{\min}, m_{\max}, \kappa) = (2, \lfloor 2T^{1/3} \rfloor, 1.0)$, and E3s uses the same bandwidth path $\{m_t\}$ as E3h.

7.8 Main Monte Carlo findings (using the reported runs)

Baselines preserve admissibility; hard truncations do not (and tapering mitigates). In the reported runs, E1 and E2 exhibit no admissibility failures in any design: $\Pr(\min_\omega \widehat{f}(\omega) < 0) = 0$ and $\Pr(\lambda_{\min}(\widehat{\Omega}) < 0) = 0$. In contrast, hard truncations generate nontrivial finite-sample failures. Under DGP2 at $T = 250$, $\Pr(\min_\omega \widehat{f}(\omega) < 0) = 0.2020$ for E0 and 0.1605 for E3h, while $\Pr(\lambda_{\min}(\widehat{\Omega}) < 0) = 0.0590$ (E0) and 0.0625 (E3h). Under DGP1 at $T = 250$, $\Pr(\min_\omega \widehat{f}(\omega) < 0) = 0.0865$ (E0) and 0.0425 (E3h), while $\Pr(\lambda_{\min}(\widehat{\Omega}) < 0) = 0.0115$ (E0) and 0.0060 (E3h). Frequencies decline sharply with T. Holding the same state rule $\{m_t\}$ fixed, the tapered state-dependent estimator E3s exhibits no such failures in the reported runs.

Inference distortions in small samples; PSD repair helps only partially. In small samples, hard truncations (E0/E3h) display undercoverage and size inflation relative to E1/E2. For example, under DGP1 at $T = 250$, coverage is 0.9050 (E0) and 0.9095 (E3h) versus 0.9260–0.9270 (E1/E2), while size is 0.0950 (E0) and 0.0905 (E3h) versus 0.0730–0.0740 (E1/E2). Under DGP2 at $T = 250$, size rises to 0.0805 (E0/E3h) from 0.0560–0.0565 (E1/E2). PSD projection reduces size in the stressed DGP2 case (e.g., from 0.0805 to 0.0765 for E0 and to 0.0785 for E3h at $T = 250$) but does not fully restore nominal behavior.

7.9 Empirical results: admissibility, severity, and inference

The figures and tables in this section are intended to be read jointly. Figure 1 summarizes the *frequency* of admissibility failures across estimators, DGPs, and sample sizes. Figure 2 and Figure 3 summarize the *severity* of the failures via the empirical distributions of $\min_\omega \widehat{f}(\omega)$ and $\lambda_{\min}(\widehat{\Omega})$. Figure 4 links these diagnostics to *downstream inference* (coverage, size, and local power). Table 1 reports corresponding frequency and severity moments (including scale normalization by $\widehat{f}(0)$), while Table 2 reports unconditional inference diagnostics together with conditional distortions *given* negativity events. Table 3 isolates the effect of PSD projection for hard-truncation estimators, and Table 4 provides a compact side-by-side comparison showing that the tapered state-dependent estimator (E3s) matches the admissibility behavior of the PSD-preserving baselines in the reported runs.

Frequency patterns (Figure 1 and Table 1). Across both DGPs, admissibility failures concentrate in the hard-truncation constructions (E0, E3h) at smaller sample sizes, and recede as T increases. The contrast between E3h and E3s is informative because they share the same state-dependent bandwidth path $\{m_t\}$: holding the state rule fixed, removing the discontinuity at the boundary (Bartlett taper) eliminates failures in the reported runs. This design comparison is consistent with the boundary-mechanism interpretation developed in the theory sections: when a cutoff is discontinuous and state-dependent, boundary-driven oscillations can be amplified in finite samples.

Severity and scale (Figure 2, Figure 3). The severity histograms highlight that failures manifest as a left tail of $\min_\omega \widehat{f}(\omega)$ and $\lambda_{\min}(\widehat{\Omega})$. At $T = 250$ the hard-truncation estimators exhibit thicker left tails and occasional crossings below zero, whereas the PSD-preserving baselines remain concentrated away from the boundary. At $T = 1000$ the left tails largely recede. Table 1 complements these histograms by reporting both the mean of the minima and a normalization $\widehat{f}_{\min}/\widehat{f}(0)$ that anchors the magnitude of failures to the estimated long-run variance scale.

Inference implications (Figure 4 and Tables 2–3). Figure 4 and Table 2 show that, in small samples, hard truncations are associated with undercoverage and size inflation relative to E1/E2. Table 2 further reports conditional distortions *given* negativity events (negative spectra or negative eigenvalues), which separates (i) the baseline finite-sample behavior of the procedure from (ii) the additional deterioration observed on the subset of replications in which admissibility fails. Table 3 then isolates the effect of PSD projection (eigenvalue clipping) for hard-truncation estimators: in the stressed designs, projection reduces size inflation but does not fully restore nominal behavior, indicating that ex post repair addresses feasibility but not the upstream source of variability induced by truncation.

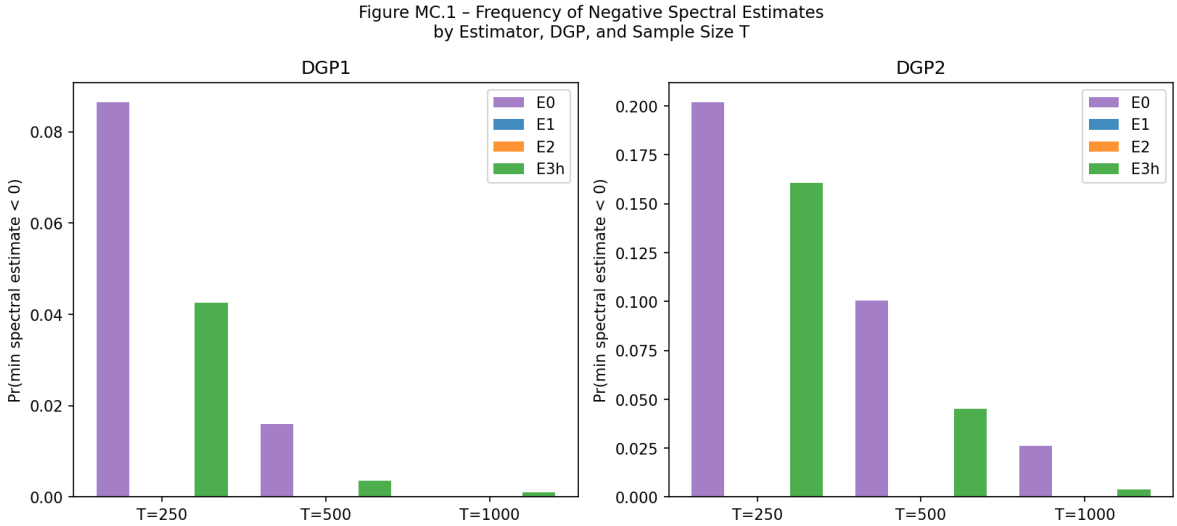

Figure 1: Frequency of negative spectral estimates. Monte Carlo frequency $\Pr(\min_{\omega\in\mathcal{G}} \widehat{f}(\omega) < 0)$ by estimator (E0, E1, E2, E3h, E3s), DGP, and sample size T. E0 is a fixed hard cutoff (rectangular truncation) control; E3h is a state-dependent hard cutoff with a moving boundary; E3s is its tapered (smooth-kernel) analogue using Bartlett weights with the same state-dependent bandwidth path.

7.10 Parameter map for (α, β) (Step 5): screening + refined robustness checks

Because sign changes (when present) can be extremely small in magnitude, we use a three-stage pipeline. **Stage 1 (screening).** On a grid in (α, β), evaluate a fast, non-certified approximation $\widetilde{W}_{\alpha,\beta}(t)$ on a coarse t-grid, record $\widetilde{m}_{\alpha,\beta} = \min_t \widetilde{W}_{\alpha,\beta}(t)$ and the minimizer $\widetilde{t}^*_{\alpha,\beta}$, and flag candidates with $\widetilde{m}_{\alpha,\beta} < -\varepsilon_{\text{scr}}$. In the reported parameter maps we set $\varepsilon_{\text{scr}} = 10^{-10}$. **Stage 2 (refined robustness checks).** For a representative subset of screened cells (most negative, boundary, and reference-neighborhood points), rerun the evaluation with tighter numerical settings (denser t grid and stricter quadrature tolerances) and label points as robust negative or not robust at

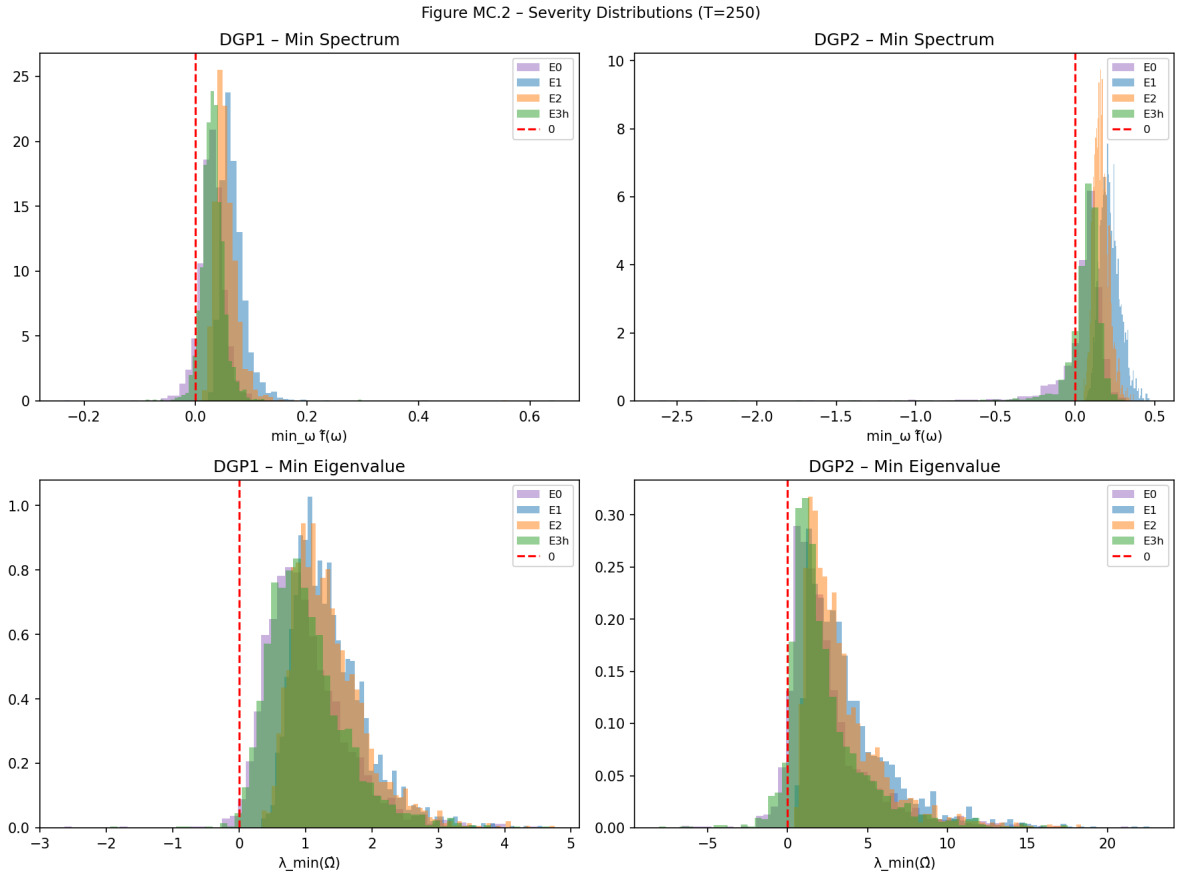

Figure 2: (T=250) – Severity distributions in small samples. Histograms of $\min_\omega \widehat{f}(\omega)$ (top row) and $\lambda_{\min}(\widehat{\Omega})$ (bottom row), by estimator and DGP. Hard truncations (E0, E3h) exhibit thicker left tails and occasional admissibility failures, while PSD-preserving baselines (E1, E2) remain concentrated away from the boundary.

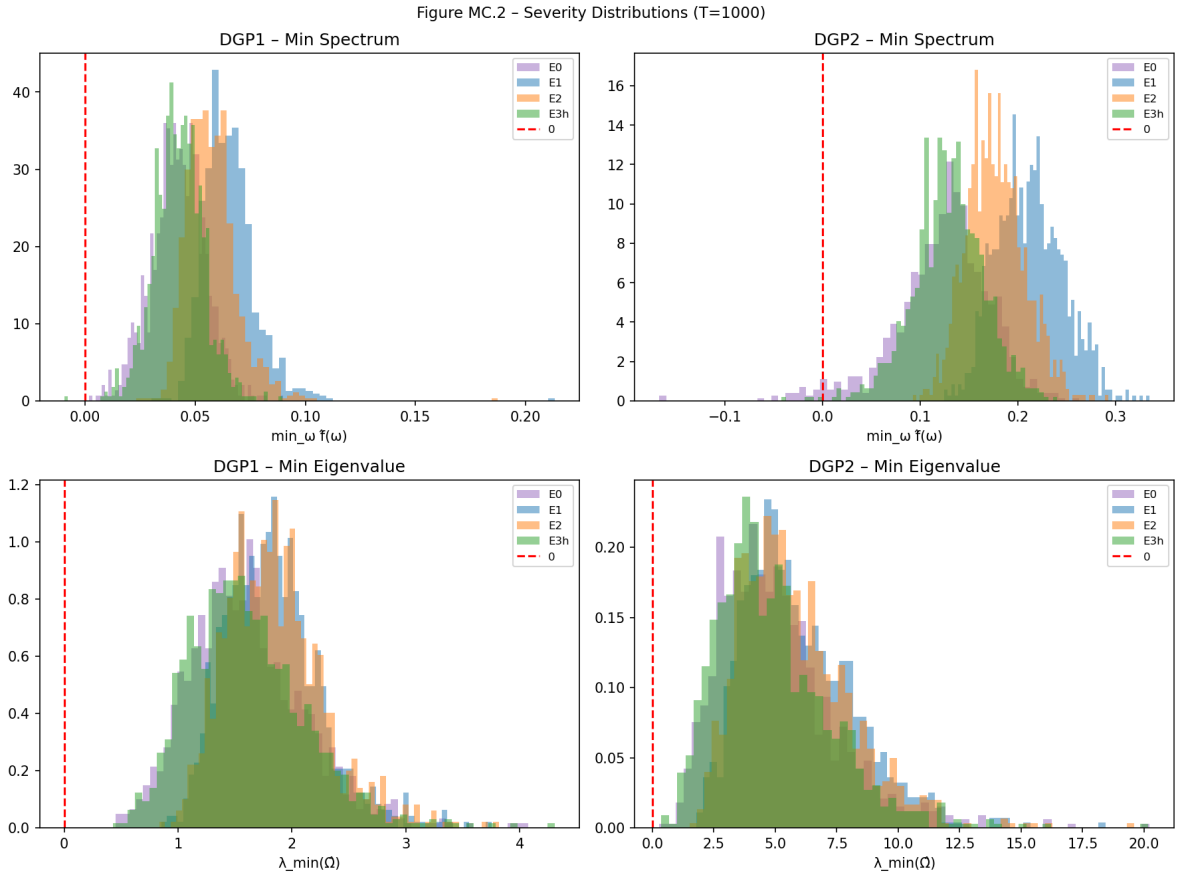

Figure 3: (T=1000) – Severity distributions in larger samples. Same diagnostics as Figure 2 but at $T = 1000$. Left tails largely recede, consistent with a finite-sample boundary effect.

Table 1: Frequency and severity of negative spectral estimates and non-PSD long-run covariance.

DGP	T	Est.	$\Pr(\hat{f}_{\min} < 0)$	$\mathbb{E}[\hat{f}_{\min}]$	$\Pr(\hat{\lambda}_{\min} < 0)$	$\mathbb{E}[\hat{\lambda}_{\min}]$	$\mathbb{E}$[neg eig]	$\mathbb{E}$[dist PSD]	$\mathbb{E}[\hat{f}(0)]$	$\mathbb{E}[\hat{f}_{\min}/\hat{f}(0)]$
DGP1	250	E0	0.0865	0.002749	0.0115	1.0140	0.0007735	0.0007735	0.05075	0.05416
DGP1	250	E1	0	0.06648	0	1.3770	0	0	0.06078	1.0940
DGP1	250	E2	0	0.06548	0	1.3550	0	0	0.06077	1.0790
DGP1	250	E3h	0.0425	0.002753	0.0060	1.0240	0.0003926	0.0003926	0.05075	0.05424
DGP1	250	E3s	0	0.06639	0	1.3690	0	0	0.06078	1.0930
DGP1	500	E0	0.0160	0.005462	0.0005	1.3700	0.00003593	0.00003593	0.03133	0.1734
DGP1	500	E1	0	0.04373	0	1.6530	0	0	0.03384	1.2920
DGP1	500	E2	0	0.04358	0	1.6500	0	0	0.03384	1.2880
DGP1	500	E3h	0.0035	0.005473	0.0010	1.3620	0.00009205	0.00009205	0.03133	0.1734
DGP1	500	E3s	0	0.04369	0	1.6390	0	0	0.03384	1.2910
DGP1	1000	E0	0	0.05337	0	1.6080	0	0	0.02005	2.6620
DGP1	1000	E1	0	0.03661	0	1.8260	0	0	0.02133	1.7160
DGP1	1000	E2	0	0.03730	0	1.8440	0	0	0.02133	1.7490
DGP1	1000	E3h	0.0010	0.05271	0	1.6030	0	0	0.02005	2.6280
DGP1	1000	E3s	0	0.03671	0	1.8150	0	0	0.02133	1.7210
DGP2	250	E0	0.2020	0.06788	0.0590	2.5150	0.007637	0.007637	0.1687	0.4097
DGP2	250	E1	0	0.1943	0	3.8800	0	0	0.1884	1.0310
DGP2	250	E2	0	0.1842	0	3.4330	0	0	0.1878	0.9808
DGP2	250	E3h	0.1605	0.06259	0.0625	2.3940	0.008150	0.008150	0.1685	0.3801
DGP2	250	E3s	0	0.1924	0	3.7620	0	0	0.1883	1.0220
DGP2	500	E0	0.1005	0.1115	0.0150	3.6920	0.001670	0.001670	0.1080	1.0250
DGP2	500	E1	0	0.2518	0	4.8830	0	0	0.1298	1.9400
DGP2	500	E2	0	0.2495	0	4.5400	0	0	0.1295	1.9270
DGP2	500	E3h	0.0450	0.1236	0.0130	3.5320	0.001446	0.001446	0.1078	1.1400
DGP2	500	E3s	0	0.2512	0	4.7380	0	0	0.1298	1.9350
DGP2	1000	E0	0.0260	0.1357	0	5.0620	0	0	0.06440	2.1050
DGP2	1000	E1	0	0.2837	0	5.9630	0	0	0.07755	3.6580
DGP2	1000	E2	0	0.2892	0	5.7910	0	0	0.07746	3.7330
DGP2	1000	E3h	0.0040	0.1508	0	4.8980	0	0	0.06424	2.3530
DGP2	1000	E3s	0	0.2824	0	5.8420	0	0	0.07754	3.6410

Table 2: Inference diagnostics (coverage, size, power) and conditional distortions when negativity occurs. Empty conditional cells correspond to zero events.

DGP	T	Est.	$\text{Cov}_{.95}$	$\text{Sz}_{.05}$	Pwr	Inadm	N_{nf}	$N_{n\lambda}$	Sz| nf	Cov| nf	Sz| $n\lambda$	Cov| $n\lambda$
DGP1	250	E0	0.9050	0.0950	0.1770	0.0015	173	23	0.2081	0.7688	0.4783	0.4783
DGP1	250	E1	0.9260	0.0740	0.1525	0	0	0	—	—	—	—
DGP1	250	E2	0.9270	0.0730	0.1495	0	0	0	—	—	—	—
DGP1	250	E3h	0.9095	0.0905	0.1745	0.0005	85	12	0.1412	0.7882	0.2500	0.5000
DGP1	250	E3s	0.9255	0.0745	0.1520	0	0	0	—	—	—	—
DGP1	500	E0	0.9295	0.0705	0.1375	0	32	1	0.1875	0.6875	1.0000	0.0000
DGP1	500	E1	0.9370	0.0630	0.1240	0	0	0	—	—	—	—
DGP1	500	E2	0.9385	0.0615	0.1215	0	0	0	—	—	—	—
DGP1	500	E3h	0.9230	0.0770	0.1360	0	7	2	0.1429	0.7143	0.0000	1.0000
DGP1	500	E3s	0.9370	0.0630	0.1240	0	0	0	—	—	—	—
DGP1	1000	E0	0.9360	0.0640	0.1260	0	0	0	—	—	—	—
DGP1	1000	E1	0.9390	0.0610	0.1120	0	0	0	—	—	—	—
DGP1	1000	E2	0.9410	0.0590	0.1090	0	0	0	—	—	—	—
DGP1	1000	E3h	0.9400	0.0600	0.1230	0	1	0	0.0000	1.0000	—	—
DGP1	1000	E3s	0.9400	0.0600	0.1110	0	0	0	—	—	—	—
DGP2	250	E0	0.9195	0.0805	0.1175	0.0055	404	118	0.1708	0.7607	0.3644	0.5508
DGP2	250	E1	0.9435	0.0565	0.0870	0	0	0	—	—	—	—
DGP2	250	E2	0.9440	0.0560	0.0865	0	0	0	—	—	—	—
DGP2	250	E3h	0.9195	0.0805	0.1190	0.0060	321	125	0.1776	0.7632	0.3440	0.5360
DGP2	250	E3s	0.9420	0.0580	0.0895	0	0	0	—	—	—	—
DGP2	500	E0	0.9325	0.0675	0.0950	0.0010	201	30	0.1294	0.7861	0.2333	0.7000
DGP2	500	E1	0.9455	0.0545	0.0785	0	0	0	—	—	—	—
DGP2	500	E2	0.9495	0.0505	0.0760	0	0	0	—	—	—	—
DGP2	500	E3h	0.9345	0.0655	0.0935	0.0015	90	26	0.1333	0.7778	0.1923	0.7308
DGP2	500	E3s	0.9470	0.0530	0.0790	0	0	0	—	—	—	—
DGP2	1000	E0	0.9570	0.0430	0.0790	0	26	0	0.1154	0.8077	—	—
DGP2	1000	E1	0.9530	0.0470	0.0700	0	0	0	—	—	—	—
DGP2	1000	E2	0.9540	0.0460	0.0690	0	0	0	—	—	—	—
DGP2	1000	E3h	0.9520	0.0480	0.0740	0	4	0	0.2500	0.2500	—	—
DGP2	1000	E3s	0.9530	0.0470	0.0690	0	0	0	—	—	—	—

Table 3: Effect of PSD projection (eigenvalue clipping) on size and coverage for hard-truncation estimators.

DGP	T	Est.	$\text{Size}_{0.05}$	$\text{Size}_{0.05}$_PSD	$\text{Coverage}_{0.95}$	$\text{Coverage}_{0.95}$_PSD	Power (local)	$\Pr(\widehat{\lambda}_{\min} < 0)$
DGP1	250	E0	0.0950	0.0945	0.9050	0.9055	0.1770	0.0115
DGP1	250	E3h	0.0905	0.0905	0.9095	0.9095	0.1745	0.0060
DGP1	500	E0	0.0705	0.0705	0.9295	0.9295	0.1375	0.0005
DGP1	500	E3h	0.0770	0.0770	0.9230	0.9230	0.1360	0.0010
DGP1	1000	E0	0.0640	0.0640	0.9360	0.9360	0.1260	0
DGP1	1000	E3h	0.0600	0.0600	0.9400	0.9400	0.1230	0
DGP2	250	E0	0.0805	0.0765	0.9195	0.9230	0.1175	0.0590
DGP2	250	E3h	0.0805	0.0785	0.9195	0.9210	0.1190	0.0625
DGP2	500	E0	0.0675	0.0665	0.9325	0.9330	0.0950	0.0150
DGP2	500	E3h	0.0655	0.0655	0.9345	0.9345	0.0935	0.0130
DGP2	1000	E0	0.0430	0.0430	0.9570	0.9570	0.0790	0
DGP2	1000	E3h	0.0480	0.0480	0.9520	0.9520	0.0740	0

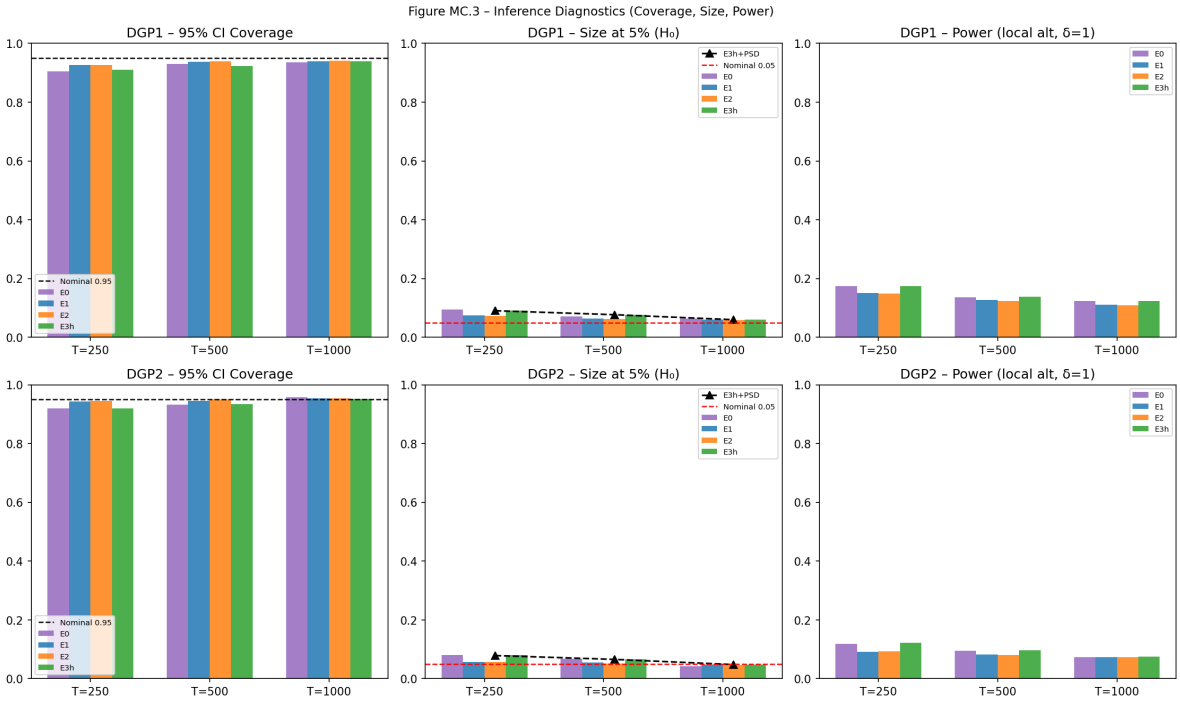

Figure 4: Inference diagnostics (coverage, size, power). Monte Carlo 95% CI coverage, size at 5% under H_0, and local power for β_1, comparing E0, E1, E2, E3h, and E3s. The PSD-repaired variant (eigenvalue clipping) is shown to distinguish admissibility failures from downstream inference effects.

Table 4: E3s comparative diagnostics (tapered moving bandwidth).

DGP	T	R	Est.	$\Pr(\hat{f}_{\min} < 0)$	$\mathbb{E}[\hat{f}_{\min}]$	$\Pr(\hat{\lambda}_{\min} < 0)$	$\mathbb{E}[\hat{\lambda}_{\min}]$	$\mathrm{Cov}_{.95}$	$\mathrm{Sz}_{.05}$	$\mathrm{Sz}_{.05}$_PSD	Pwr	$\mathbb{E}[\mathrm{neg}\,\lambda]$
DGP1	250	2000	E1	0.0000	0.06648	0.0000	1.3773	0.9260	0.0740	0.0740	0.1525	0.0000
DGP1	250	2000	E3h	0.0425	0.002753	0.0060	1.0238	0.9095	0.0905	0.0905	0.1745	0.0003926
DGP1	250	2000	E3s	0.0000	0.06639	0.0000	1.3689	0.9255	0.0745	0.0745	0.1520	0.0000
DGP1	500	2000	E1	0.0000	0.04373	0.0000	1.6529	0.9370	0.0630	0.0630	0.1240	0.0000
DGP1	500	2000	E3h	0.0035	0.005473	0.0010	1.3621	0.9230	0.0770	0.0770	0.1360	0.00009205
DGP1	500	2000	E3s	0.0000	0.04369	0.0000	1.6395	0.9370	0.0630	0.0630	0.1240	0.0000
DGP1	1000	1000	E1	0.0000	0.03661	0.0000	1.8255	0.9390	0.0610	0.0610	0.1120	0.0000
DGP1	1000	1000	E3h	0.0010	0.05271	0.0000	1.6035	0.9400	0.0600	0.0600	0.1230	0.0000
DGP1	1000	1000	E3s	0.0000	0.03671	0.0000	1.8148	0.9400	0.0600	0.0600	0.1110	0.0000
DGP2	250	2000	E1	0.0000	0.1943	0.0000	3.8798	0.9435	0.0565	0.0565	0.0870	0.0000
DGP2	250	2000	E3h	0.1605	0.06259	0.0625	2.3941	0.9195	0.0805	0.0785	0.1190	0.008150
DGP2	250	2000	E3s	0.0000	0.1924	0.0000	3.7625	0.9420	0.0580	0.0580	0.0895	0.0000
DGP2	500	2000	E1	0.0000	0.2518	0.0000	4.8833	0.9455	0.0545	0.0545	0.0785	0.0000
DGP2	500	2000	E3h	0.0450	0.1236	0.0130	3.5322	0.9345	0.0655	0.0655	0.0935	0.001446
DGP2	500	2000	E3s	0.0000	0.2512	0.0000	4.7381	0.9470	0.0530	0.0530	0.0790	0.0000
DGP2	1000	1000	E1	0.0000	0.2837	0.0000	5.9630	0.9530	0.0470	0.0470	0.0700	0.0000
DGP2	1000	1000	E3h	0.0040	0.1508	0.0000	4.8983	0.9520	0.0480	0.0480	0.0740	0.0000
DGP2	1000	1000	E3s	0.0000	0.2824	0.0000	5.8419	0.9530	0.0470	0.0470	0.0690	0.0000

that numerical resolution. Unless explicitly stated otherwise, these remain refined checks (not certified computation). **Stage 3 (sparse certification, Appendix A standard).** For 5–10 representative points selected from the map (near the reference neighborhood, strong-negative region, boundary region, and positive controls), we certify the sign at a candidate t_0 using the explicit truncation/quadrature error budget developed in Appendix A: we construct $\widetilde{W}_{\alpha,\beta}(t_0)$ together with a bound $|\widetilde{W}_{\alpha,\beta}(t_0) - W_{\alpha,\beta}(t_0)| \leq \varepsilon_{\text{tot}}$ and declare *NEGATIVE CERTIFIED* when $\widetilde{W}_{\alpha,\beta}(t_0) + \varepsilon_{\text{tot}} < 0$.

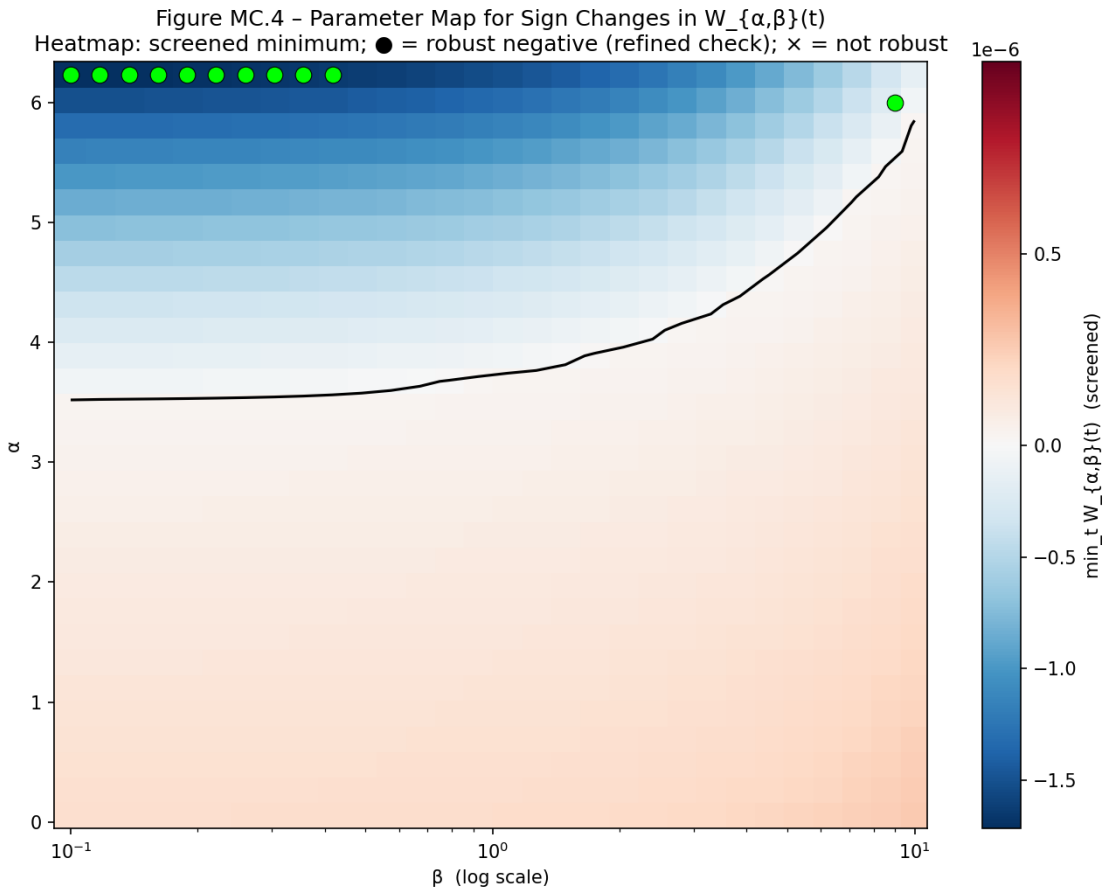

Figure 5: Parameter map for sign changes in $W_{\alpha,\beta}(t)$. Heatmap of screened minima $\min_t \widetilde{W}_{\alpha,\beta}(t)$ over an (α, β) grid, with markers indicating points that pass a refined numerical robustness check.

7.11 Honest scope statement for the Monte Carlo section

These experiments are not meant to establish a universal theorem about HAC failure under all forms of adaptivity. Rather, they quantify: (i) how often and how severely non-admissibility occurs under hard truncations (E0) and a mechanism-targeted moving truncation (E3h) relative to standard baselines (E1–E2); (ii) whether PSD projection reduces size distortion in stressed designs; and (iii) how the sign-change region for the analytic family $H_{\alpha,\beta}$ behaves empirically under a transparent screening + refined-check protocol, with sparse certification at selected points. Appendix A provides pointwise validated certification at selected t_0 values.

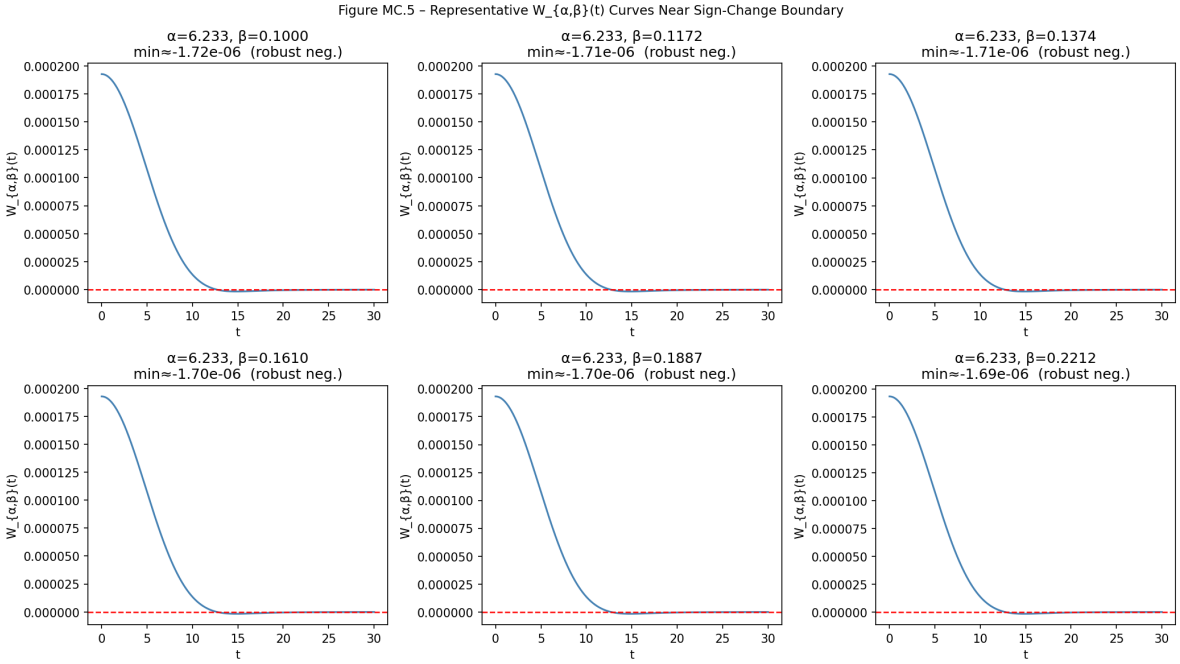

Figure 6: Representative $W_{\alpha,\beta}(t)$ curves near the empirical sign-change boundary. Examples of $W_{\alpha,\beta}(t)$ for selected (α, β) pairs near the boundary identified in Figure 5, illustrating small-magnitude negative dips consistent with the screening and refined-check pipeline.

8 Discussion and concluding remarks

For our explicit family, the boundary density $\rho_{\alpha,\beta}$ is available in closed form (see (25)), and Lemma 5.7 shows that the single object controlling the sign of W is the defect density $\kappa = -H'$. Proposition 6.3 therefore reduces global nonnegativity of W to a single sine–transform positivity condition. Proposition 6.5 provides an explicit nontrivial test case in which this mechanism can be verified in closed form.

On the econometric side, the Monte Carlo section documents that admissibility failures (negative spectral estimates and non–PSD HAC matrices) concentrate in hard-cutoff implementations, particularly when the cutoff index is state-dependent, while the tapered state-dependent variant (E3s) exhibits no such failures in the reported designs. Taken together, the analytic results and the simulations support a coherent interpretation: moving hard boundaries can introduce truncation-driven oscillatory terms that may overturn naive positivity intuition, whereas smooth tapering mitigates boundary ringing. Importantly, the theory is presented as a diagnostic mechanism in a fully tractable analogue rather than as a universal theorem for all adaptive HAC procedures, and the Monte Carlo evidence is correspondingly scoped to the stated DGPs, tuning rules, and reported experiments.

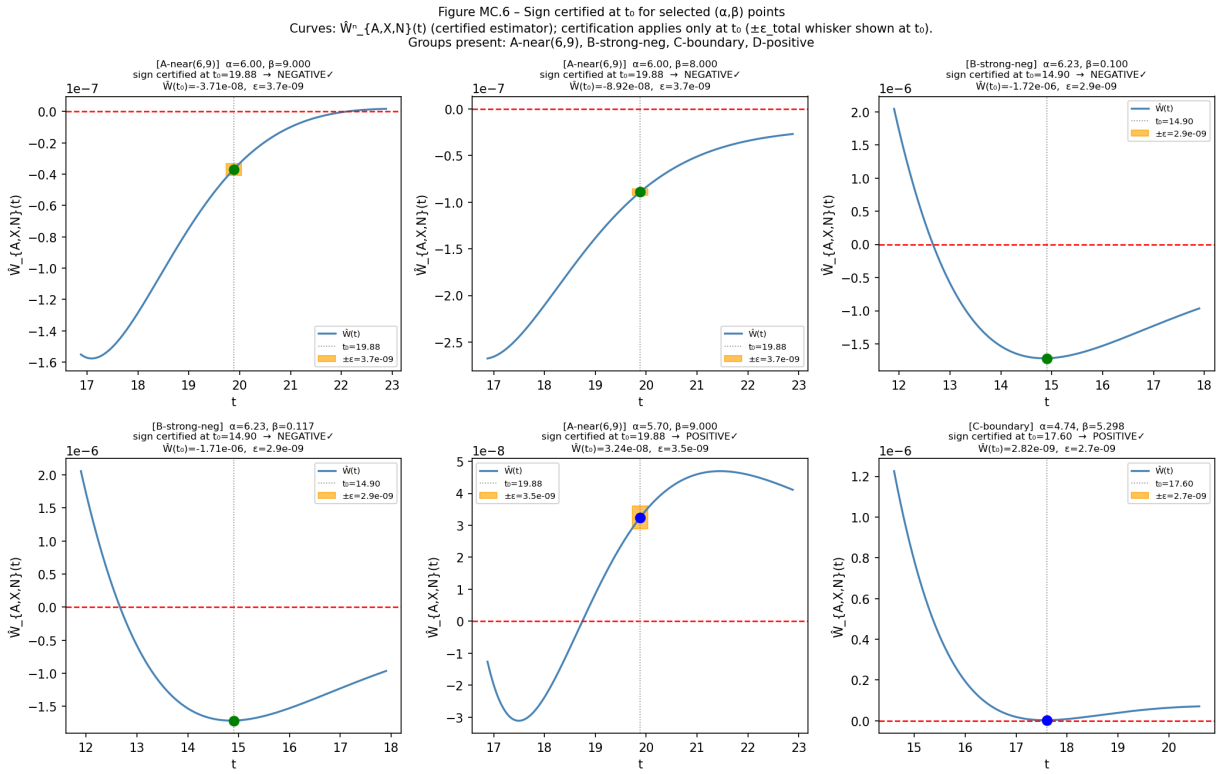

Figure 7: Sign certified at t_0 for selected (α, β) points. Curves $\widetilde{W}_{\alpha,\beta}(t)$ are shown for representative points across the screened map (near the reference neighborhood, strong-negative region, boundary region, and positive controls). Certification applies only at the marked t_0 (whisker indicates $\pm\varepsilon_{\text{tot}}$), yielding a rigorous sign conclusion for $W_{\alpha,\beta}(t_0)$.

Appendix A. Validated numerics for a certified sign reversal of the cosine–transform functional

Appendix A documents validated-numerics certification (explicit truncation and quadrature error bounds) for the reference parameters $(\alpha, \beta) = (6, 9)$ and for a sparse set of additional representative points drawn from the parameter map Moore et al., 2009; Rump, 1999; Tucker, 2011. Accordingly, we write $H(x) = H_{6,9}(x)$ and $W(t) = W_{6,9}(t)$.

$$W_{6,9}(t) = \int_{-\infty}^{\infty} H_{6,9}(x) \cos(tx)\, dx,$$

where $H_{6,9}$ is defined in (8)–(10). The purpose of this appendix is not to provide a qualitative illustration, but to establish a fully reproducible inequality of the form $W_{6,9}(t_0) < 0$ with explicit truncation and quadrature error bounds. This certification complements the analytic mechanism in the main text: the strict pointwise positivity of $H_{6,9}(x)$ established in Theorem 4.2 and the bookkeeping in Remark 4.1 does not preclude boundary-induced oscillatory cancellations in the cosine transform. By Corollary 3.7, once a negative value is certified at a fixed frequency for the reference parameters, the sign-change phenomenon persists under small perturbations of (α, β).

Our numerical strategy is as follows. We approximate the integral defining $H_{6,9}(x)$ by truncating the outer a–integration at a cutoff A, exploiting the super–exponential decay of $e^{-a \cosh x}$. We then approximate the cosine transform by truncating the x–integration to $[-X, X]$ and applying a deterministic quadrature rule on $[0, X]$ (symmetry). The validated step is to bound the total discrepancy between the true value $W(t_0)$ and the computed approximation $\widetilde{W}_{A,X,N}(t_0)$ by an explicit error budget, so that a strict inequality $\widetilde{W}_{A,X,N}(t_0) + \varepsilon_{\text{tot}} < 0$ implies $W(t_0) < 0$.

Certified sign change (validated numerics). To rule out numerical artifacts, we use the validated error bound in Lemma 5.3. For the reference case $(\alpha, \beta) = (6, 9)$ we fix

$$X = 3, \qquad A = 200, \qquad N = 131072 \quad (h = X/N),$$

and evaluate $\widetilde{W}_{A,X,N}(t)$ with the trapezoidal rule in x (outer quadrature), using the exact integrand definition of H_A. The numerical evaluation yields

$$\widetilde{W}_{A,X,N}(19.8798) \approx -3.705979 \times 10^{-8}.$$

Moreover, the quantities M_0, M_1, M_2 and $B_\partial(X)$ appearing in Lemma 5.3 can be computed from one-dimensional integrals together with a maximization over $u \in [0, X]$ of explicit boundary expressions, which gives the bound

$$\left| W(19.8798) - \widetilde{W}_{A,X,N}(19.8798) \right| \leq 2.1 \times 10^{-9}.$$

Since $2.1 \times 10^{-9} < 3.7 \times 10^{-8}$, it follows rigorously that $W(19.8798) < 0$. In other words, the negative value at $t_0 = 19.8798$ is certified by an explicit error budget and is not attributable to domain truncation or quadrature error at the stated numerical settings.

Figures A.1–A.2 display non-certified numerical approximations of $W_{6,9}(t)$ over a range of frequencies, intended only to visualize the qualitative shape of the transform. Certification in this appendix is pointwise: the rigorous conclusion concerns the sign of $W_{6,9}(t_0)$ at the reported t_0 values (Table A.1), not the sign of the entire curve.

In particular, the numerical approximation yields values such as

$$W_{6,9}(15.79) \approx -8.3 \times 10^{-9}, \qquad W_{6,9}(19.88) \approx -3.7 \times 10^{-8},$$

Table A.1: Multi-point sign certification at selected (α, β) points. Certification is pointwise at the reported t_0 with total error bound $\varepsilon_{\text{total}}$. A sign is declared certified when $\widehat{W}(t_0) + \varepsilon_{\text{total}} < 0$ (NEG) or $\widehat{W}(t_0) - \varepsilon_{\text{total}} > 0$ (POS). Boundary points are near-threshold by construction.

group	alpha	beta	t0	W_hat	eps_A	eps_X	eps_trap	eps_inner	eps_total	neg_cert	pos_cert	certified	sign	verdict	M0	M1	M2
Bpartial	C2																
A-near(6,9)	6	9	19.88	-3.706e-08	8.309e-10	2.537e-09	2.739e-11	3.145e-10	3.692e-09	True	False	True	NEG	CERTIFIED	0.0001521	0.0002016	0.0002769
0.0001134	1.56																
A-near(6,9)	5.7	9	19.88	3.245e-08	7.993e-10	2.447e-09	2.677e-11	2.735e-10	3.55e-09	False	True	True	POS	CERTIFIED	0.0001322	0.0001748	0.0002404
0.00009817	1.56																
A-near(6,9)	6	8	19.88	-8.917e-08	8.021e-10	2.519e-09	2.743e-11	3.102e-10	3.658e-09	True	False	True	NEG	CERTIFIED	0.0001462	0.0001937	0.0002664
0.0001101	1.56																
B-strong-neg	6.233	0.1	14.9	-1.717e-06	6.874e-10	1.897e-09	3.338e-11	2.9e-10	2.91e-09	True	False	True	NEG	CERTIFIED	0.0002741	0.0004041	0.0006356
0.0001678	1.56																
B-strong-neg	6.233	0.1172	14.9	-1.714e-06	7.271e-10	1.861e-09	3.337e-11	2.886e-10	2.91e-09	True	False	True	NEG	CERTIFIED	0.0002699	0.0003991	0.0006294
0.0001678	1.56																
C-boundary	4.741	5.298	17.6	2.816e-09	5.967e-10	1.822e-09	2.76e-11	2.89e-10	2.74e-09	False	True	True	POS	CERTIFIED	0.0001119	0.0001472	0.0002014
0.00007916	1.56																
C-boundary	4.954	6.21	17.6	3.49e-09	6.557e-10	1.889e-09	2.764e-11	2.875e-10	2.86e-09	False	True	True	POS	CERTIFIED	0.0001175	0.0001547	0.0002127
0.00008312	1.56																
D-positive	0.2632	8.532	30.9	2.122e-07	4.712e-10	5.236e-10	1.698e-11	2.887e-10	1.3e-09	False	True	True	POS	CERTIFIED	0.00001801	0.00002252	0.00002872
0.00001047	1.56																
D-positive	0.4764	8.532	30.9	2.042e-07	5.383e-10	6.018e-10	1.698e-11	3.008e-10	1.46e-09	False	True	True	POS	CERTIFIED	0.00002433	0.0000305	0.00003922
0.00001396	1.56																

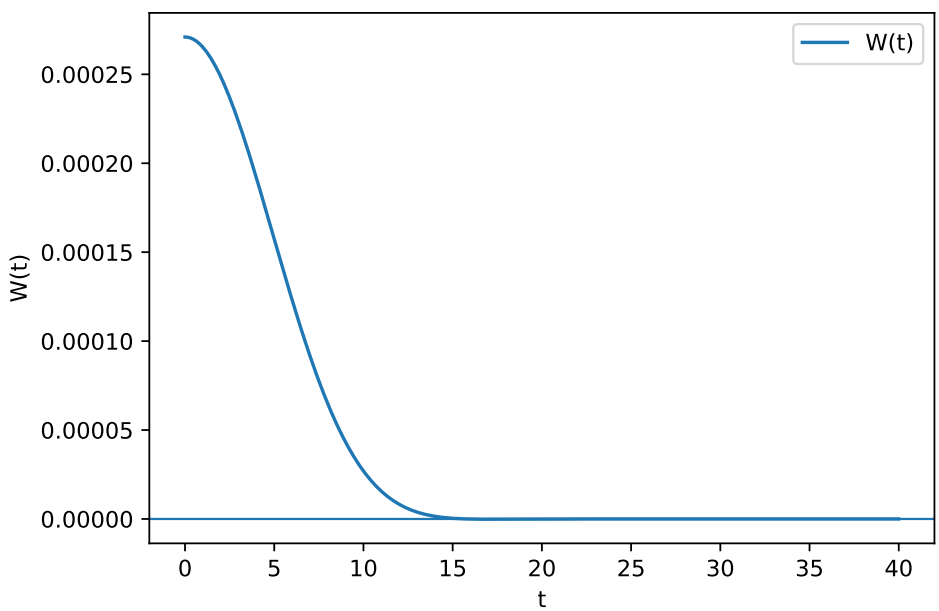

Figure A.1: Global numerical approximation of the cosine–transform functional $W_{6,9}(t)$. The curve is not certified pointwise except at the reported t_0 values.

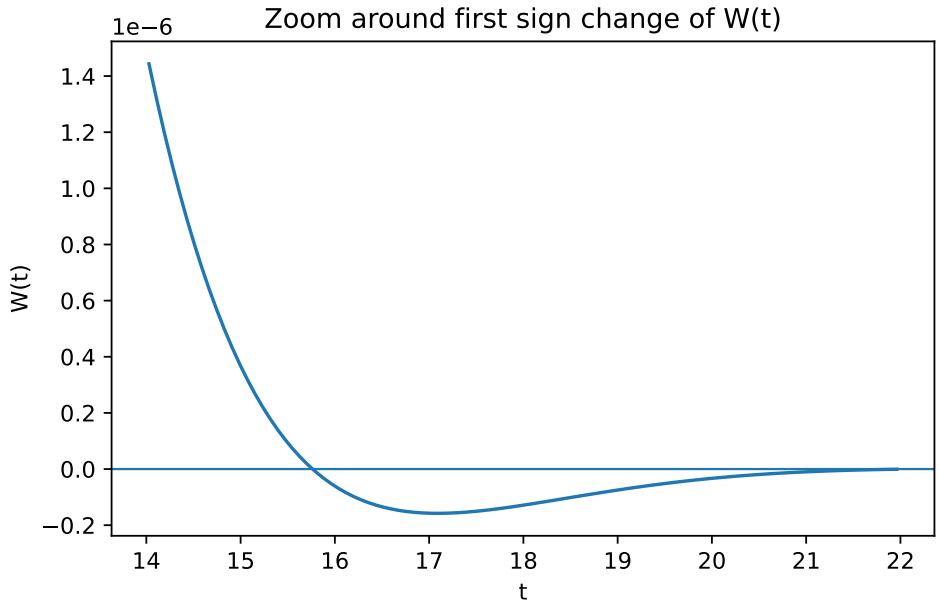

Figure A.2: Zoom around a certified sign change of $W_{6,9}(t)$ near $t_0 = 19.8798$. The negative sign is certified only at the marked frequency(ies), not uniformly over the plotted interval.

which are consistent with the certified inequality at $t_0 = 19.8798$ and illustrate the small magnitude scale at which sign reversals can occur for this reference kernel.

References

Andrews, D. W. K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, *59*(3), 817–858. https://doi.org/10.2307/2938229

Andrews, D. W. K., & Monahan, J. C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, *60*(4), 953–966. https://doi.org/10.2307/2951574

Bartlett, M. S. (1950). Periodogram analysis and continuous spectra. *Biometrika*, *37*(1-2), 1–16. https://doi.org/10.1093/biomet/37.1-2.1

Brockwell, P. J., & Davis, R. A. (1991). *Time series: Theory and methods* (2nd ed.). Springer.

Hardy, G. H. (1949). *Divergent series*. Clarendon Press.

Higham, N. J. (2002). Computing the nearest correlation matrix—a problem from finance. *IMA Journal of Numerical Analysis*, *22*(3), 329–343. https://doi.org/10.1093/imanum/22.3.329

Katznelson, Y. (2004). *An introduction to harmonic analysis* (3rd ed.). Cambridge University Press.

Kiefer, N. M., & Vogelsang, T. J. (2002). Heteroskedasticity-autocorrelation robust standard errors using the Bartlett kernel without truncation. *Econometrica*, *70*(5), 2093–2095. https://doi.org/10.1111/1468-0262.00366

Kiefer, N. M., & Vogelsang, T. J. (2005). A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. *Econometric Theory*, *21*(6), 1130–1164. https://doi.org/10.1017/S0266466605050565

Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, *88*(2), 365–411. https://doi.org/10.1016/S0047-259X(03)00096-4

Moore, R. E., Kearfott, R. B., & Cloud, M. J. (2009). *Introduction to interval analysis*. SIAM.

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610

Newey, W. K., & West, K. D. (1994). Automatic lag selection in covariance matrix estimation. *The Review of Economic Studies*, *61*(4), 631–653. https://doi.org/10.2307/2297912

Phillips, P. C. B., Sun, Y., & Jin, S. (2006). Spectral density estimation and robust hypothesis testing using steep origin kernels without truncation. *International Economic Review*, *47*(3), 837–894. https://doi.org/10.1111/j.1468-2354.2006.00398.x

Politis, D. N. (2011). Higher-order accurate, positive semidefinite estimation of large-sample covariance and spectral density matrices. *Econometric Theory*, *27*(4), 703–744. https://doi.org/10.1017/S0266466610000484

Priestley, M. B. (1981). *Spectral analysis and time series*. Academic Press.

Rudin, W. (1987). *Real and complex analysis* (3rd ed.). McGraw-Hill.

Rump, S. M. (1999). INTLAB—INTerval LABoratory. In T. Csendes (Ed.), *Developments in reliable computing* (pp. 77–104). Kluwer Academic Publishers.

Tucker, W. (2011). *Validated numerics: A short introduction to rigorous computations*. Princeton University Press.